



NETWORKING
WITH ISRAR

LEARN | PRACTICE | SUCCEED

SUBNETTING

/24 vs **/25** vs **/26** vs **/27** vs **/28**

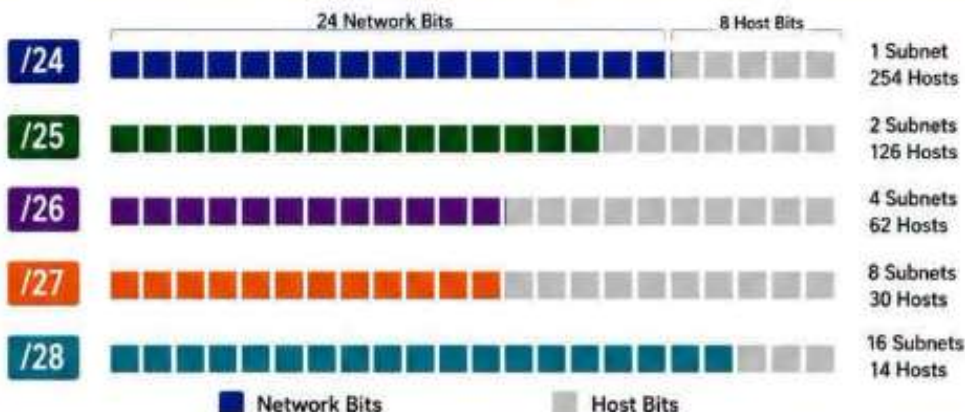


Understand Subnets. Plan Better. Design Smarter Networks.

	/24 255.255.255.0	/25 255.255.255.128	/26 255.255.255.192	/27 255.255.255.224	/28 255.255.255.240
Subnet Mask (Binary)	11111111.11111111. 11111111.00000000 24 8	11111111.11111111. 11111111.10000000 25 7	11111111.11111111. 11111111.11000000 26 6	11111111.11111111. 11111111.11100000 27 5	11111111.11111111. 11111111.11110000 28 4
CIDR Notation	/24	/25	/26	/27	/28
Total Addresses Per Subnet	256	128	64	32	16
Usable Hosts Per Subnet	254	126	62	30	14
Network Increment	256	128	64	32	16
Example Subnets (Last Octet)	192.168.1.0/24 (One Network)	0, 128	0, 64, 128, 192	0, 32, 64, 96, 128, 160, 192, 224	0, 16, 32, 48, 64, 80, 96, 112, 128, 144, 160, 176, 192, 208, 224, 240
Usable Host Range (Example) (192.168.1.x)	192.168.1.1 – 192.168.1.254	192.168.1.1 – 192.168.1.126	192.168.1.1 – 192.168.1.62	192.168.1.1 – 192.168.1.30	192.168.1.1 – 192.168.1.14
Broadcast Address (Example)	192.168.1.255	192.168.1.127	192.168.1.63	192.168.1.31	192.168.1.15



SUBNET SIZE VISUALIZATION



WHY SUBNET?



Efficient use of
IP addresses



Better network
organization



Improved network
performance



Enhanced security
and control



Easier to scale
your network

50Mbps Link at 95%. Voice Unaffected. QoS Is the Policy That Makes It Happen.

SD-WAN QoS QUEUING & BANDWIDTH ALLOCATION



CONGESTION SCENARIO (LINK AT 95% UTILIZATION)

- Link utilization reaches 95%
 - Backup transfer fills Queue 4 (Scavenger)
 - Queue 4 packets dropped first
 - Queue 0 (Voice) remains unaffected
 - Voice quality maintained MOS Score 4.2+
- QoS protects what matters. Voice gets priority.**

DSCP CLASSIFICATION & MARKING

DSCP	PHB	QUEUE	APPLICATION EXAMPLES
EF (46)	Expedited Forwarding	Q0 (Voice)	Voice, VoIP, IP Phone Signaling, SRTP
AF41 (34)	Assured Forwarding 41	Q1 (Video)	Video Conferencing, Webex, Zoom, Teams
AF31 (26)	Assured Forwarding 31	Q2 (Business)	SAP, Oracle, Salesforce, Office 365, CRM
AF21 (18)	Assured Forwarding 21	Q3 (General)	Web Browsing, Email, Social Media, Updates
CS1 (8)	Class Selector 1	Q4 (Scavenger)	Backup, File Transfer, Bulk Data Transfer
BE (0)	Best Effort	Q4 (Scavenger)	Default Traffic, Unclassified Traffic

VMANAGE QoS MAP TEMPLATE

QoS Map: BRANCH-TO-MPLS-QOS-MAP				
Queue ID	Queue Name	Bandwidth (Mbps)	Percent	Type
0	Voice	10	20%	Strict Priority
1	Video	12.5	25%	WRR
2	Business-Apps	10	20%	WRR
3	General	12.5	25%	WRR
4	Scavenger	5	10%	Scavenger

Additional Settings

- Default Queue: 4 (Scavenger)
- DSCP to Queue Map: Enabled
- Traffic Shaping: Enabled (50 Mbps)
- Tail Drop: Enabled on Queue 4

Template Attached to All Branch Devices

REAL WORLD IMPACT: VOICE QUALITY COMPARISON

BEFORE QoS (NO PRIORITIZATION)

Link Congested (95% Utilization)



- All traffic competes equally
- Voice packets experience delay & jitter
- Packet loss occurs across all queues
- Users experience choppy audio / dropped calls

MOS SCORE

2.8

☆☆☆☆☆

UNACCEPTABLE

VS

AFTER QoS (WITH PRIORITIZATION)

Link Congested (95% Utilization)



- Voice (Q0) gets strict priority
- Other queues share bandwidth fairly
- Scavenger (Q4) drops first
- Voice remains clear and uninterrupted

MOS SCORE

4.2

☆☆☆☆☆

EXCELLENT

KEY TAKEAWAY

QoS ensures the right application gets the right performance, every time.



Protect Voice Quality



Better Video Experience



Efficient Bandwidth Utilization



Business App Performance



Lower Costs & Higher Productivity



POLICY + PRIORITY = PREDICTABLE PERFORMANCE



**NETWORKING
WITH ISRAR**

NETWORKING BASICS



HOW DATA MOVES THROUGH NETWORKS

1 OSI MODEL (7 LAYERS)

Layer	Function	Examples
7 Application	User Interface	HTTP, DNS, SMTP
6 Presentation	Data Format	Encryption, JPEG
5 Session	Session Control	NetBIOS, RPC
4 Transport	End-to-End	TCP, UDP
3 Network	Routing	IP, ICMP
2 Data Link	Framing, MAC	Ethernet, MAC
1 Physical	Cables, Signals	Cables, Hubs



FLOW OF DATA

- 1 Application creates data
- 2 Goes down the OSI layers
- 3 Sent over the network
- 4 Received & goes up layers

2 TCP/IP MODEL

Layer	Protocols
Application (Layer 5-7)	HTTP, DNS, FTP
Transport (Layer 4)	TCP, UDP
Internet (Layer 3)	IP
Network Access (Layer 1-2)	Ethernet, Wi-Fi

Mapping to OSI

- Application → Layers 5-7
- Transport → Layer 4
- Internet → Layer 3
- Network Access → Layers 1-2

3 IP ADDRESSING (IPv4)

- Format : 192.168.1.10
- Public vs Private
- Private Ranges :
 - ✓ 10.0.0.0 / 8
 - ✓ 172.16.0.0 / 12
 - ✓ 192.168.0.0 / 16
- Loopback : 127.0.0.1
- IPv6 Example : 2001:db8::1



4 SUBNETTING

Example: 192.168.1.0/24

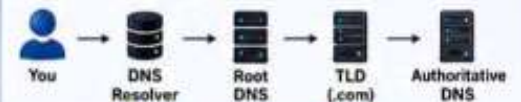


- ✓ Network Address : 192.168.1.0
- ✓ Broadcast Address : 192.168.1.255
- ✓ Usable Hosts : 192.168.1.1 – 192.168.1.254

TIP: More 1s in subnet mask = Fewer Hosts

5 DNS (DOMAIN NAME SYSTEM)

- Translates domain names to IP addresses
- Example: google.com → 142.250.x.x



Common Records

- A → IPv4 Address
- AAAA → IPv6 Address
- CNAME → Alias
- MX → Mail Server

6 DHCP (Dynamic Host Config.)



Automatically provides:

- ✓ IP Address
- ✓ Subnet Mask
- ✓ Default Gateway
- ✓ DNS Server

DORA Process

- 1 Discover
- 2 Offer
- 3 Request
- 4 Acknowledge



7 NAT (Network Address Translation)

- Allows private devices to access the Internet
- Translates Private IP to Public IP



Done by your Router / Firewall

8 COMMON PROTOCOLS

- TCP → Reliable, Connection-Oriented
- UDP → Fast, Connectionless
- ICMP → Ping, Diagnostics
- DNS → Domain Name System (UDP/TCP 53)
- DHCP → IP Assignment (UDP 67/68)
- HTTP → Web Traffic (TCP 80)
- HTTPS → Secure Web (TCP 443)



9 WIRESHARK BASICS

- Tool to capture & analyze network traffic

Steps:

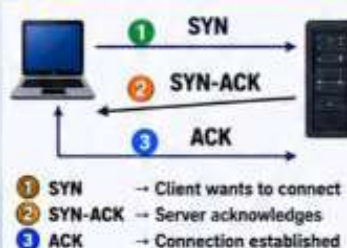
- ✓ Select Interface
- ✓ Capture Traffic
- ✓ Analyze Packets
- ✓ Filter Results



Filters Examples

ip.addr == 192.168.1.10 | tcp | dns | http

10 TCP 3-WAY HANDSHAKE



11 DATA FLOW EXAMPLE (Visiting a Website)

- ✓ Open Browser
- ✓ DNS Resolves Domain
- ✓ TCP Connection (3-Way)
- ✓ HTTP/HTTPS Request
- ✓ Server Response
- ✓ Data Displayed



12 KEY TAKEAWAYS

- ✓ Understand OSI & TCP/IP
- ✓ Know IP Addressing & Subnetting
- ✓ Understand DNS, DHCP & NAT
- ✓ Use Wireshark to Analyze Traffic
- ✓ Follow how Data Travels Step by Step



NETWORKING WITH ISRAR
LEARN | PRACTICE | SUCCEED



GEMINI PRO

KERJA, BELAJAR,
RISET JADI
LEBIH MUDAH

AI canggih dari Google
yang siap membantumu
setiap hari!



FITUR GEMINI PRO

-  **Gemini 1.5 Pro**
Model AI terbaru, lebih pintar & akurat
-  **Deep Research**
Riset mendalam dalam hitungan menit
-  **Upload & Analisis Dokumen**
PDF, Docs, Sheets, Slides & lainnya
-  **NotebookLM Plus**
Rangkum, pahami & kelola informasi
-  **Buat Gambar & Visual**
Hasil visual berkualitas tinggi
-  **Bantuan Coding**
Dari ide hingga eksekusi

AMAN, CEPAT & TERPERCAYA

-  Akun Google Pribadi
-  Aktivasi Cepat
-  Bergaransi
-  Support Responsif

 Bantu riset lebih cepat & mendalam

 Brainstorming ide tanpa batas

 Menulis, merangkum & membuat konten lebih mudah

 Bantu coding & debugging lebih efisien

AKSES PREMIUM, HARGA TERJANGKAU!

HARGA NORMAL
~~Rp 225.000~~

HANYA
Rp 75.000

/TAHUN

HEMAT
66%

<

FILTER

HALAMAN

LOKASI ▼

KATEGORI ▼

**Inci Sözlük Jailbreak · Ikuti**

Game/Mainan · 1 pengikut

**IPHONE Jailbreak · Ikuti**Landmark · 5 Manor Dr, Apt 4A, Newark,
New Jersey, New Jersey, Amerika Serikat ·
17 pengikut

Berdasarkan pencarian Anda untuk 'jailbreak ai'

**Nexora Ai**

Bersponsor ·

GEMINI PRO

KERJA, BELAJAR,
RISET JADI
LEBIH MUDAH

AI canggih dari Google
yang siap membantumu
setiap hari!

- Bantu riset lebih cepat & mendalam
- Brainstorming ide tanpa batas
- Menulis, merangkum & membuat konten lebih mudah
- Bantu coding & debugging lebih efisien

AKSES PREMIUM, HARGA TERJANGKAU!

HARGA NORMAL
~~Rp 225.000~~

HANYA
Rp 75.000
/TAHUN

HEMAT 66%

FITUR GEMINI PRO

- Gemini 1.5 Pro**
Model AI terbaru, lebih pintar & akurat
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Dari ide hingga eksekusi

AMAN, CEPAT & TERPERCAYA

- Akun Google Pribadi
- Aktivasi Cepat
- Bergaransi
- Support Responsif

Gemini Pro Mulai Rp75.000

Ingin menggunakan Gemini Pro untuk

10/10



CyberKid
LEARN. SECURE. DEFEND.



EXPERIENCE LEVEL:

MASTER THE FUTURE

ELITE CERTIFICATIONS. LIMITLESS IMPACT.

These advanced certifications validate your expertise, strategic thinking and leadership to build a safer digital world.



PERFECT FOR YOU IF:

- You lead or want to lead security teams
- You design strategy and architecture
- You drive risk management and governance
- You want to make a global impact

1

(ISC)[®]
CISSP[®]

(Certified Information Systems
Security Professional)

LEVEL: EXPERT

The most recognized security certification worldwide. Covers architecture, engineering, risk, governance and more.



2

ISACA[®]
CISM[®]

(Certified Information
Security Manager)

LEVEL: EXPERT

Designed for security managers. Focuses on governance, risk management and program development.



3

EC-Council
C|CISO[®]

(Certified Chief Information
Security Officer)

LEVEL: EXPERT

Executive-level certification for aspiring CISOs and security executives. Focuses on strategy, leadership and business.



4

TOGAF[®]
Enterprise Architecture
(Security Focus)

(The Open Group)

LEVEL: EXPERT

Enterprise architecture framework to design secure, scalable and resilient organizations.



5

aws

AWS Certified Security –
Specialty

(SCS-C02)

LEVEL: ADVANCED

Validates your ability to design and implement secure AWS solutions at scale.



CORE EXPERTISE & LEADERSHIP AREAS YOU WILL MASTER



Security
Architecture



Risk Management
& Governance



Enterprise
Strategy



Leadership
& Communication



Cloud & Enterprise
Security



Threat Intelligence
& Analysis



Compliance
& Assurance



Innovation
& Resilience



KNOWLEDGE BUILDS SKILLS.
SKILLS BUILD SECURITY.



STRATEGY DRIVES SUCCESS.
LEADERSHIP CREATES IMPACT.



PROTECT TODAY.
SECURE TOMORROW.



BE THE EXPERT.
INSPIRE OTHERS.

9/10



CyberKid
LEARN. SECURE. DEFEND.



EXPERIENCE LEVEL:

SPECIALIST & ARCHITECT

LEAD THE FUTURE OF SECURITY

You're ready to become a strategist,
design secure systems and lead
organizations into a safer digital future.



PERFECT FOR YOU IF:

- ✓ You have deep security knowledge & experience
- ✓ You want to design strategies & secure architectures
- ✓ You aim for leadership & executive roles
- ✓ You want to make a global impact



LEADERSHIP & STRATEGIC SKILLS YOU WILL MASTER

 Security Architecture & Design	 Risk Management & Governance	 Leadership & Communication	 Strategy & Business Alignment	 Cloud & Enterprise Security	 Compliance & Assurance	 Innovation & Transformation	 Global Impact & Resilience
--	--	--	---	---	--	---	--

1

(ISC)[®]
CISSP[®]

(Certified Information Systems
Security Professional)

LEVEL: EXPERT

The gold standard for security professionals. Covers architecture, engineering, risk, governance and more.



2

TOGAF[®]

Enterprise Architecture
(Security Focus)

(The Open Group)

LEVEL: EXPERT

Enterprise architecture framework to design, plan and implement secure, scalable and resilient organizations.



3

CISA[®]

(Certified Information
Systems Auditor)

LEVEL: EXPERT

Focuses on auditing, control and assurance of information systems to manage risk and ensure compliance.



4

aws

Advanced Networking –
Specialty

(SCS-C01)

LEVEL: ADVANCED

Design and implement complex networking solutions on AWS with a focus on security, scale and performance.



5

A Microsoft Certified
Azure Solutions
Architect Expert

(AZ-305)

LEVEL: EXPERT

Design secure, scalable and high-availability solutions on Microsoft Azure and drive digital transformation.



YOUR KNOWLEDGE
BUILDS SECURITY.



YOUR STRATEGY
DRIVES SUCCESS.



YOUR LEADERSHIP
CREATES IMPACT.

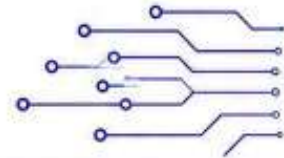


YOUR CERTIFICATIONS
OPEN DOORS.



YOUR FUTURE
HAS NO LIMITS.

8/10


CyberKid
 LEARN. SECURE. DEFEND.


EXPERIENCE LEVEL:

CLOUD & MULTI-CLOUD

SECURE THE FUTURE

Cloud skills are in high demand. These certifications validate your ability to **secure**, **architect** and **manage** cloud environments.



PERFECT FOR YOU IF:

- ✓ You work with cloud platforms (AWS, Azure, GCP)
- ✓ You want to become a Cloud Security Engineer
- ✓ You aim for high-paying cloud security roles
- ✓ You want to future-proof your career



1

aws

 AWS Certified
 Security – Specialty
 (SCS-C02)

LEVEL: ADVANCED

Proves your expertise in securing AWS workloads and building secure cloud solutions.



2

Microsoft
Certified
 Azure Security
 Engineer Associate
 (AZ-500)

LEVEL: ADVANCED

Validate your ability to implement security controls and protect data, identities and cloud resources.



3

Google Cloud
Professional
 Cloud Security
 Engineer
 (Professional)

LEVEL: ADVANCED

Demonstrates skills in securing Google Cloud solutions and managing risk and compliance.



4

(ISC)²
CCSP®
 (Certified Cloud
 Security Professional)

LEVEL: ADVANCED

Global certification for cloud security professionals. Covers architecture, governance, risk and compliance.



5

CompTIA
Cloud+

(CV0-003)

LEVEL: INTERMEDIATE

Vendor-neutral cert covering cloud concepts, architecture, operations, security and troubleshooting.



TOP CLOUD SKILLS YOU WILL MASTER


 Cloud Security
 & Governance

 Identity & Access
 Management

 Secure Architecture
 & Design

 Data Protection
 & Encryption

 Monitoring &
 Threat Detection

 Incident Response
 in the Cloud

 Compliance &
 Risk Management

 DevSecOps
 & Automation

 THE CLOUD IS THE FUTURE.
 SECURITY IS THE ADVANTAGE.

 UPSKILL TODAY.
 LEAD TOMORROW.

 SECURE EVERY CLOUD.
 PROTECT EVERY FUTURE.

 YOUR SKILLS.
 YOUR POWER.
 YOUR FUTURE.

7/10



CyberKid
LEARN. SECURE. DEFEND.



EXPERIENCE LEVEL:

EXPERT

MASTER, LEAD, INNOVATE

You're at the top of your game. These elite certifications validate your mastery, leadership and ability to solve complex real-world security challenges.



PERFECT FOR YOU IF:

- ✓ You lead or want to lead security teams
- ✓ You design security architecture & strategy
- ✓ You handle risk, governance & compliance
- ✓ You aim to become a CISO or consultant



1

(ISC)²
CISSP[®]

(Certified Information Systems Security Professional)

LEVEL: EXPERT

Industry-leading certification for security leaders covering architecture, management, risk, and governance.



2

ISACA
CISM[®]

(Certified Information Security Manager)

LEVEL: EXPERT

Designed for security managers. Focuses on governance, risk management and program development.



3

TOGAF[®]

Enterprise Architecture (Security Focus)

(The Open Group)

LEVEL: EXPERT

Enterprise architecture framework to design secure, scalable and resilient organizations.



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EC-Council
C|CISO[®]

(Certified Chief Information Security Officer)

LEVEL: EXPERT

Exec-level certification for aspiring CISOs and security executives. Focus on strategy, leadership and business.



5

SANS
MGT512

(Cybersecurity Leadership Essentials)

LEVEL: EXPERT

Build leadership skills to manage security teams, communicate effectively and drive security programs.



EXPERTISE & LEADERSHIP AREAS



Security Strategy & Leadership



Risk Management & Governance



Enterprise Architecture



Team Leadership & Communication



Compliance & Legal



Innovation & Research



Business Alignment & Impact



MASTERY IS EARNING.
IMPACT IS LEGACY.



LEAD TODAY.
INSPIRE TOMORROW.



STRATEGY. PEOPLE.
SECURITY.

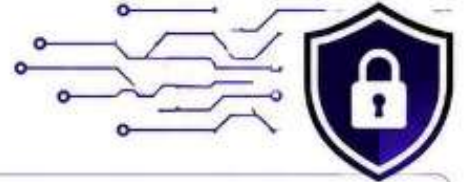


BE THE LEADER.
SHAPE THE FUTURE.

6/10



CyberKid
LEARN. SECURE. DEFEND.



EXPERIENCE LEVEL:

ADVANCED

BECOME A SECURITY EXPERT

You're ready for the next level.
These certifications prove **deep expertise**,
leadership ability and advanced
security knowledge.



PERFECT FOR YOU IF:

- ✓ You have strong hands-on experience
- ✓ You want to lead teams or projects
- ✓ You aim for senior or specialist roles
- ✓ You want to make a bigger impact



1

(ISC)²
CISSP[®]

(Certified Information Systems
Security Professional)

LEVEL: ADVANCED

Global standard for security
leaders. Covers security
architecture, management,
risk and governance.



2

OffSec
OSCP

(Offensive Security
Certified Professional)

LEVEL: ADVANCED

Hands-on certification that
proves advanced penetration
testing skills in real-world
labs.



3

CREST
CRTO

(Certified Red Team
Operator)

LEVEL: ADVANCED

Advanced red teaming
certification focused on
adversary emulation and
attack simulations.



4

GIAC
GXPN

(GIAC Exploit Researcher
and Advanced Penetration Tester)

LEVEL: ADVANCED

Advanced exploitation and
penetration testing cert
for experienced security
professionals.



5

ISACA
CISM[®]

(Certified Information
Security Manager)

LEVEL: ADVANCED

Designed for security
managers. Focuses on
governance, risk management
and program development.



EXPERTISE AREAS YOU WILL MASTER



Security
Architecture



Risk Management
& Governance



Advanced
Penetration Testing



Red Team
Operations



Threat Intelligence
& Analysis



Cloud & Enterprise
Security



Security Leadership
& Strategy



EXPERIENCE BUILDS SKILLS.
EXPERTISE CREATES IMPACT.



INVEST IN YOURSELF.
LEAD THE FUTURE.



BE THE EXPERT
OTHERS RELY ON.



ADVANCED TODAY.
STRONGER TOMORROW.

5/10



CyberKid
LEARN. SECURE. DEFEND.



EXPERIENCE LEVEL:

INTERMEDIATE+

DEEPEN YOUR EXPERTISE

You've gained solid skills.
Now go even deeper, *specialize*
in advanced areas and prepare
for high-demand roles.



PERFECT FOR YOU IF:

- ✓ You have 1-2 years of cybersecurity experience
- ✓ You want to specialize in offensive or defensive security
- ✓ You aim for high-level roles and better salaries
- ✓ You are ready for advanced challenges



1

OffSec
OSCP

(Offensive Security
Certified Professional)

LEVEL: ADVANCED

Industry-respected offensive security cert that proves your penetration testing skills in real-world scenarios.



2

CREST
CRTO

(Certified Red Team
Operator)

LEVEL: ADVANCED

Hands-on red teaming certification that validates your ability to simulate real-world attacks.



3

(ISC)²
CISSP

(Certified Information
Systems Security Professional)

LEVEL: ADVANCED

Gold standard for security professionals and a must-have for security managers and architects.



4

Cloud Security
Alliance
CCSP

(Certified Cloud Security
Professional)

LEVEL: ADVANCED

Proves your expertise in securing cloud environments and managing cloud security risks.



5

SANS
GXPN

(GIAC Exploit Researcher
and Advanced Penetration Tester)

LEVEL: ADVANCED

Advanced certification for exploit development and advanced penetration testing techniques.



KEY SKILLS YOU WILL MASTER



Advanced Penetration
Testing



Exploit Development
& Research



Threat Hunting
& Analysis



Security Architecture
& Engineering



Cloud Security
& Compliance



Red Teaming
& Simulation



Risk Management
& Governance



PUSH YOUR LIMITS.
ACHIEVE MORE.



ADVANCED SKILLS.
BETTER OPPORTUNITIES.



BE THE EXPERT.
BE THE DIFFERENCE.



KEEP LEARNING.
NEVER STOP GROWING.

4/10



CyberKid
LEARN. SECURE. DEFEND.



EXPERIENCE LEVEL:

INTERMEDIATE

BUILD REAL-WORLD SKILLS

You have the basics.
Now it's time to **specialize**,
practice, and gain the skills
employers are looking for.



PERFECT FOR YOU IF:

- ✓ You have basic cybersecurity knowledge
- ✓ You want hands-on, practical skills
- ✓ You are preparing for mid-level roles
- ✓ You want to stand out in the job market



1

CompTIA
CySA+
(CS0-003)

LEVEL: INTERMEDIATE

Focuses on threat detection, analysis, incident response and continuous security monitoring.



2

EC-Council
CEH v13
(312-50)

LEVEL: INTERMEDIATE

Learn to think like a hacker to better understand how attacks happen and how to defend.



3

TCM Security
PNPT
(Practical)

LEVEL: INTERMEDIATE

Hands-on penetration testing certification with real-world labs and practical exams.



4

Hack The Box
HTB CDSA

(Certified Defensive Security Analyst)

LEVEL: INTERMEDIATE

Blue team focused cert teaching defense, monitoring, threat hunting and incident response.



5

GIAC
GSEC

(Security Essentials)

LEVEL: INTERMEDIATE

Vendor-neutral certification covering essential security technologies and defensive skills.



KEY SKILLS YOU WILL BUILD



Threat Detection
& Analysis



Penetration
Testing



Security Monitoring
& Logging



Risk Assessment
& Management



Incident Response
& Handling



Network & System
Security



Compliance &
Governance



SKILLS TODAY.
SECURE TOMORROW.



INVEST IN KNOWLEDGE.
GET AHEAD.



DEFEND SYSTEMS.
PROTECT PEOPLE.



KEEP LEARNING.
NEVER STOP GROWING.

3/10



CyberKid
LEARN. SECURE. DEFEND.



EXPERIENCE LEVEL:

BEGINNER+

LEVEL UP YOUR SKILLS

You've learned the basics.
Now it's time to go deeper,
build real skills and get
ready for the next step
in your cybersecurity career.



PERFECT FOR YOU IF:

- ✓ You understand the basics
- ✓ You want hands-on skills
- ✓ You're ready to go deeper
- ✓ You aim for your first job

1

CompTIA
CySA+
(CS0-003)

LEVEL: INTERMEDIATE

Focuses on threat detection, analysis, incident response and continuous security monitoring.



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CEH v13
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LEVEL: INTERMEDIATE

Hands-on penetration testing certification with real-world labs and practical exams.



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Hack The Box
HTB CDSA
(Certified Defensive Security Analyst)

LEVEL: INTERMEDIATE

Blue team focused cert teaching defense, monitoring, threat hunting and incident response.



5

Microsoft
SC-200
(Security Operations Analyst)

LEVEL: INTERMEDIATE

Build skills in security monitoring, threat hunting and using Microsoft security tools.



WHAT YOU'LL LEARN



Threat Detection
& Analysis



Incident
Response



Log Analysis
& Monitoring



Network
Security



Access Control
& IAM



Security Tools
& Automation



Reporting &
Documentation



MORE SKILLS.
MORE OPPORTUNITIES.



BUILD EXPERIENCE.
BOOST YOUR CAREER.

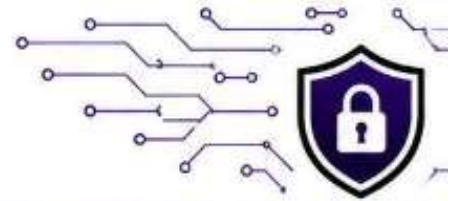


STRONGER TODAY.
SAFER TOMORROW.



KEEP LEARNING.
KEEP GROWING.

2/10

**CyberKid**
LEARN. SECURE. DEFEND.

EXPERIENCE LEVEL:

BEGINNER

START YOUR CYBERSECURITY JOURNEY

New to cybersecurity? These certifications will help you build a **strong foundation**, understand the basics and open the door to exciting opportunities.

**PERFECT FOR YOU IF:**

- ✓ You're new to cybersecurity
- ✓ You want to switch careers
- ✓ You want to build a strong foundation



1

**CompTIA
Security+**
(SY0-701)
LEVEL: ENTRY

Perfect starting point to learn core security concepts, threats, risk management and security operations.



2

eJPT
iNE
LEVEL: ENTRY

Hands-on certification that teaches practical penetration testing skills in real-world environments.



3

**Google
Cybersecurity
Certificate**
LEVEL: ENTRY

Beginner-friendly program covering security fundamentals, tools, SIEM, Python and more. No prior experience needed.



4


**CISCO
CCST Cybersecurity**
LEVEL: ENTRY

Learn cybersecurity basics, network security and threat detection with Cisco's foundational cert.



5

**(ISC)²
Certified in Cybersecurity
(CC)**
LEVEL: ENTRY

Great entry-level cert that introduces you to security principles and industry best practices.

**FOCUS AREAS YOU WILL LEARN**Security
FundamentalsThreats &
Vulnerabilities

Network Basics



Access Control

Tools &
Technologies

Best Practices

EVERY EXPERT
WAS ONCE A **BEGINNER.**LEARN TODAY.
DEFEND TOMORROW.YOUR JOURNEY STARTS
WITH THE **FIRST STEP.**

1/10



CyberKid
LEARN. SECURE. DEFEND.



BEST CYBERSECURITY CERTIFICATIONS

《《 BY EXPERIENCE LEVEL 》》

★ 2026 ★

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cybersecurity certifications at every level.
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CISSP®
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CEH
EC-Council v13

CCSP
(ISC)²

aws
Security Specialty

Azure
Security Engineer
Associate



BEGINNER



BEGINNER+



INTERMEDIATE



INTERMEDIATE+



ADVANCED



EXPERT



CLOUD CERTIFICATIONS



BLUE TEAM / RED TEAM



LEARNING ROADMAP



KNOWLEDGE TODAY
PROTECTION TOMORROW



VALIDATE YOUR SKILLS
BOOST YOUR CAREER



STAY AHEAD
STAY SECURE



GLOBAL RECOGNITION
REAL OPPORTUNITIES

WHAT WAS THE FIRST BUG?

The very first "bug" in computing history



DATE:

September 9, 1947



PLACE:

Harvard University,
Cambridge, Massachusetts, USA

THE STORY

While working on the Harvard Mark II computer, a problem was found that was causing the machine to malfunction intermittently.

Engineers discovered that a **real moth** was **trapped inside a relay**, causing the failure.

They removed the moth and taped it in the logbook with the note:



Grace Hopper

Pioneer Programmer
who popularized the term "bug"

THE ACTUAL FIRST BUG

9/9 0800 Started Cosine Tape (Sine check).
1000 Started Multi-Adder Test.
1100 Proved test. 3:05
1525 Started Cosine Tape (Sine check).
1545 Found relay # 70 Panel F (moth) in relay. 3:20

First actual case bug case of bug.

The logbook entry that recorded the **first bug**.

WHY IS IT CALLED A "BUG"?

The term "bug" was already being used in the early 1900s to describe mysterious errors.

Grace Hopper helped popularize it in computing after this incident.

“

"A moth was causing an error, so we called it a bug."

– Grace Hopper

”



THE MACHINE

Harvard Mark II

An electro-mechanical computer built in 1944. It was huge, complex, and used thousands of relays and switches.



IMPACT

- ✓ This incident became a legendary moment in computer history.
- ✓ It showed that even the most advanced machines could be affected by the smallest thing!
- ✓ It gave birth to the term "debugging" – the process of finding and removing bugs.

FUN FACTS

- 🦋 The moth was carefully removed and preserved.
- 📖 The taped moth is still preserved at the Museum of Science, Boston.
- ★ Grace Hopper is also credited with inventing the first compiler.



The original logbook
(Museum of Science, Boston)

TIMELINE OF THE FIRST BUG



1944

Harvard Mark II is built.



Sept 9, 1947

The moth is found causing the failure.



Sept 9, 1947

The event is logged as the "first actual case of bug".



1950s

The term "bug" becomes popular in computing.



Today

"Debugging" is an essential part of software development.



EVEN A **SMALL BUG** CAN CAUSE A **BIG PROBLEM**.
THAT'S WHY CAREFUL CODING AND TESTING MATTER!



Remember:

Great software is built by finding bugs and learning from them.

BUGS

Small in Size, Big in Impact!

A BUG IS

an error, flaw or unexpected behavior in a program that causes it to produce wrong or unwanted results.



WHY DO BUGS MATTER?

Bugs can crash applications, steal data, waste time, lose money and damage reputation.



System Crashes



Security Vulnerabilities



Financial Loss



Wasted Time



Bad User Experience

TYPES OF BUGS



Syntax Bugs

Mistakes in writing code that cause errors.



Logical Bugs

The program runs but doesn't work as expected.



Runtime Bugs

Errors that occur while the program is running.



Security Bugs

Flaws that attackers can exploit to harm the system.



UI/Functional Bugs

Issues in the user interface or features.

COMMON CAUSES



Incomplete Requirements



Poor Code Quality



Lack of Testing



Changes in Environment



Human Mistakes

BUG LIFE CYCLE



DISCOVERED

A bug is found by a tester, user or developer.



REPORTED

The bug is documented with details.



TRIAGED

The team analyzes and prioritizes the bug.



FIXED

Developers fix the bug in the code.



TESTED & VERIFIED

Testers verify and ensure the bug is resolved.

EXAMPLES OF BUGS IN REAL LIFE



Unable to place an order on a website



Login fails even with correct password



App crashes suddenly



Incorrect calculation results



Page freezes or loads forever

HOW TO PREVENT BUGS?



Write Clean & Readable Code



Understand Requirements Clearly



App Regularly (Unit, Integration, System Testing)



Code Review & Pair Programming



Use Automated Testing Tools



Keep Software Updated

“Bugs are not failure, ignoring them is.”



GOOD DEVELOPERS WRITE CODE. GREAT DEVELOPERS FIND AND FIX BUGS.

Stay Curious.

Keep Testing. Keep Improving.



HOW A CYBER SECURITY TEAM WORKS

"We don't just protect systems, we protect trust, data, and people."

OUR MISSION

- ✓ Protect Systems and Networks
- ✓ Detect Threats Quickly
- ✓ Respond Effectively
- ✓ Recover and Improve
- ✓ Stay One Step Ahead of Attackers

OUR GOAL

- ✓ Prevent Attacks
- 🔍 Detect Early
- ⚡ Respond Fast
- 🛡️ Minimize Damage
- 📈 Improve Security Every Day

HOW THE PROCESS WORKS

1 MONITOR



We continuously monitor systems, networks, and user activities.

2 DETECT



Identify suspicious activities or potential security threats.

3 ANALYZE



Investigate the threat, understand how it happened and its possible impact.

4 RESPOND



Take action to contain the threat and stop further damage.

5 RECOVER



Restore normal operations and ensure all systems are safe.

6 IMPROVE



Learn from the incident and strengthen security for the future.

KEY ROLES IN A CYBER SECURITY TEAM

SOC ANALYST (Front Line)



- ✓ Monitor Alerts
- ✓ Analyze Logs
- ✓ Detect Threats
- ✓ Report Incidents

THREAT HUNTER (Proactive)



- ✓ Search for Hidden Threats
- ✓ Hunt Advanced Attacks
- ✓ Use Intelligence and Tools

INCIDENT RESPONDER (Problem Solver)



- ✓ Investigate Incidents
- ✓ Contain the Threat
- ✓ Remove the Cause
- ✓ Recover Systems

SECURITY ENGINEER (Builder)



- ✓ Build Security Tools
- ✓ Configure Firewalls
- ✓ Secure Systems
- ✓ Automate Defense

FORENSIC ANALYST (Investigator)



- ✓ Collect Evidence
- ✓ Analyze Attacks
- ✓ Find Root Cause
- ✓ Support Legal Action

COMPLIANCE OFFICER (Policy & Law)



- ✓ Ensure Compliance
- ✓ Manage Policies
- ✓ Risk Assessment
- ✓ Audit & Reporting

TOOLS THEY USE EVERY DAY



Wireshark



Kali Linux



Nmap



SIEM
(Splunk/ELK)



Snort / Suricata



Burp Suite



VirusTotal



Wi-Fi / Network
Analyzers



COMMON THREATS THEY FACE



Phishing Attacks



Data Breaches



Malware / Ransomware



Insider Threats



DDoS Attacks



Social Engineering

A DAY IN THE LIFE OF A CYBER SECURITY TEAM

- 07:00 AM** Team Briefing
Review alerts, overnight reports, and priorities.
- 09:00 AM** Monitoring & Analysis
Check dashboards, logs, and security alerts.
- 12:00 PM** Threat Investigation
Analyze incidents, hunt threats, test systems.
- 03:00 PM** Incident Response / Fix
Resolve issues, patch systems, update rules.
- 06:00 PM** Documentation & Improvement
Report, learn, and prepare for tomorrow.

CYBER SECURITY IS NOT JUST A JOB, IT'S A RESPONSIBILITY.

- ✓ Think like an attacker.
- ✓ Act like a defender.
- ✓ Stay updated.
- ✓ Never stop learning.
- ✓ Protect today, secure tomorrow.

TIPS FOR A STRONG DEFENSE

- 🔑 Use Strong Passwords
- 📱 Enable Two-Factor Authentication
- 🔄 Keep Software Updated
- ⚠️ Be Careful with Links & Attachments
- 📢 Report Suspicious Activity

CYBER SECURITY IS A TEAM EFFORT — **TECHNOLOGY** + **PEOPLE** + **PROCESS** = TRUE PROTECTION

Stay Safe. Stay Aware. Stay Secure.

IPV4 SUBNETTING

SMARTER NETWORKS, BETTER PERFORMANCE



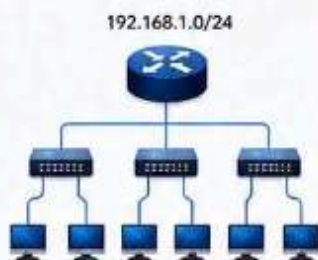
WHAT IS SUBNETTING?



IPv4 Subnetting is the process of dividing a large network into multiple smaller subnetworks. It helps to improve performance, security and efficient use of IP addresses.

WHY USE SUBNETTING?

- ✓ Reduce broadcast traffic
- ✓ Improve network performance
- ✓ Better security & segmentation
- ✓ Efficient use of IP addresses
- ✓ Easier network management



KEY TERMS

- IP** IP Address
A unique 32-bit address assigned to a device.
- Subnet Mask**
Defines the network and host portions of an IP address.
- CIDR**
Classless Inter-Domain Routing notation (e.g. /24, /26).
- Network Address**
The first address in a subnet. Represents the network.
- Broadcast Address**
The last address in a subnet. Used to reach all hosts.
- Usable Host Range**
The range of IP addresses available for devices.

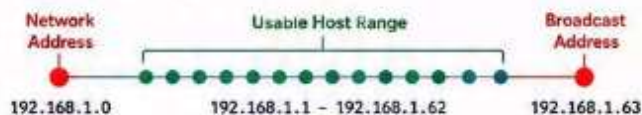
COMMON SUBNET MASKS

CIDR	SUBNET MASK	NETWORK BITS	HOST BITS	USABLE HOSTS PER SUBNET
/24	255.255.255.0	24	8	254
/25	255.255.255.128	25	7	126
/26	255.255.255.192	26	6	62
/27	255.255.255.224	27	5	30
/28	255.255.255.240	28	4	14
/29	255.255.255.248	29	3	6
/30	255.255.255.252	30	2	2

SUBNETTING EXAMPLE

Network: 192.168.1.0/26

- Subnet Mask : 255.255.255.192
- CIDR : /26
- Network Address : 192.168.1.0
- First Usable Host : 192.168.1.1
- Last Usable Host : 192.168.1.62
- Broadcast Address : 192.168.1.63
- Total Addresses : 64
- Usable Hosts : 62



HOW SUBNETTING WORKS



SUBNETTING FORMULAS

Number of Subnets	Usable Hosts per Subnet
2^n	$2^h - 2$
n = Number of borrowed host bits	h = Number of remaining host bits
TIP First IP = Network Address (Not Usable) Last IP = Broadcast Address (Not Usable)	

SUBNETTING PRACTICE (EXAMPLE)

Network	CIDR	Subnet Mask	Usable Host Range	Broadcast
192.168.1.0	/24	255.255.255.0	192.168.1.1 – 192.168.1.254	192.168.1.255
192.168.1.0	/25	255.255.255.128	192.168.1.1 – 192.168.1.126 192.168.1.129 – 192.168.1.254	192.168.1.127 192.168.1.255
192.168.1.0	/26	255.255.255.192	192.168.1.1 – 192.168.1.62 192.168.1.64 – 192.168.1.126 192.168.1.128 – 192.168.1.190 192.168.1.192 – 192.168.1.254	192.168.1.63 192.168.1.127 192.168.1.191 192.168.1.255
10.0.0.0	/30	255.255.255.252	10.0.0.1 – 10.0.0.2	10.0.0.3

QUICK REFERENCE



Subnetting makes networks efficient and scalable.



Better performance
+ Better security
+ Better management



Practice more,
Understand deeply,
Build your skills.



Master Subnetting,
Master Networking!

TCP/IP MODEL

Transmission Control Protocol/Internet Protocol

TCP/IP condenses the seven layers of the OSI model into four. It simplifies the upper and lower layers because, in practice, the functions often overlap. The OSI model is a theoretical framework used for teaching and standardization, while TCP/IP is a functional stack that internet actually runs on.

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4



Application Layer

combines the functions of the OSI Model's Application, Presentation, and Session layers. Purpose: Provides an interface between the user and the network.

HTTP/HTTPS, DNS, SMTP/IMAP, SSH/Telnet.

Data unit: Data or message

3



Transport Layer

manages the end-to-end communication between the source and destination hosts. It is responsible for data integrity and flow control. Purpose: Ensures that data reaches the correct application on a device using Ports.

TCP, UDP. Data unit: Segments(for TCP),
Datagram(for UDP)

2



Internet Layer

handles the movement of data across multiple networks. Responsible for logical addressing and finding the most efficient path for data to travel from the source to the destination.

IP, ICMP, ARP. Data unit: Packets

1



Link Layer (Network Access Layer)

handles the physical and electrical components of data transmission. Purpose: Defines how data is physically sent through the medium, such as fiber optics, copper wire, or radio waves. It manages hardware addressing (MAC addresses).

Ethernet, Wi-Fi, Fiber optics. Data unit:
Frames(converted to bits)

What is ICMP?



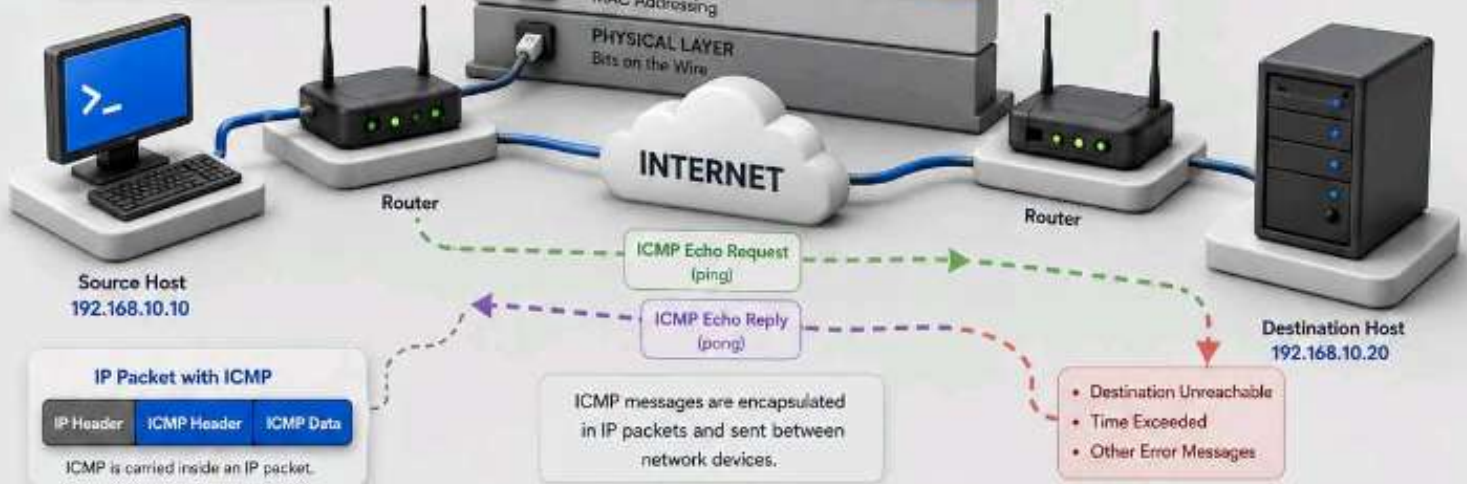
ICMP (Internet Control Message Protocol) is a network layer protocol used by devices to send **error messages** and **operational information** about IP network communications.

Common ICMP Messages

- Echo Request / Reply (ping)
- Destination Unreachable
- Time Exceeded
- Redirect
- Source Quench (deprecated)

What ICMP Does

- Reports errors
- Tests connectivity
- Sends operational info
- Helps troubleshoot network issues



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ICMP Use Cases

Ping

Echo Request/Reply
Tests if a host is reachable and measures round-trip time.



Error Reporting
Notifies hosts when a packet cannot be delivered (e.g., host unreachable).



Path Diagnostics
Helps tools like traceroute identify network path issues.



Network Management
Provides operational info for better network performance.



Troubleshooting
Essential for network admins to quickly identify and resolve issues.

HOME VLAN SETUP MAP

Separate Homelab, IoT & Main Devices on ONE router

No enterprise gear





FIREWALL TERMS



— Key Concepts Every Network Engineer Should Know —

FIREWALL FUNDAMENTALS



Firewall

A security device that monitors and controls incoming and outgoing network traffic based on rules.



Packet Filtering

Examines packets and allows or blocks them based on source, destination, IP, port and protocol.



Stateful Inspection

Tracks the state of active connections and makes decisions based on the context of the traffic.



Proxy Firewall

Acts as an intermediary between users and the internet, hiding internal IPs and filtering traffic at the application layer.



NGFW (Next-Gen Firewall)

Advanced firewall with deep packet inspection, IPS, app control and user awareness.



UTM (Unified Threat Management)

Combines multiple security features like firewall, antivirus, IPS, VPN and more in one solution.

FIREWALL TYPES



Network Firewall

Protects the entire network by controlling traffic at the perimeter.



Host-based Firewall

Runs on individual devices to control incoming and outgoing traffic.



Cloud Firewall

Firewall service delivered from the cloud to protect cloud resources.



Web Application Firewall (WAF)

Protects web applications by filtering HTTP/HTTPS traffic.



Firewall as a Service (FWaaS)

Cloud-based firewall service that scales with your infrastructure.



Virtual Firewall

Software-based firewall that runs in virtual environments.

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KEY FEATURES



Access Control

Allows or blocks traffic based on predefined security rules.



NAT (Network Address Translation)

Hides internal IP addresses by translating them to a public IP.



DPI (Deep Packet Inspection)

Inspects packet payloads to detect threats and enforce policies.



IPS/IDS (Intrusion Prevention/Detection)

Detects and blocks malicious activities or alerts on suspicious behavior.



Application Control

Controls and filters traffic based on applications and their usage.



URL Filtering

Blocks or allows website access based on categories or policies.

COMMON FIREWALL TERMS



Rule / Policy

A set of conditions and actions that determine how traffic is handled.



Logical grouping of interfaces with similar security levels.



DMZ (Demilitarized Zone)

A network segment that exposes public services to untrusted networks.



Allow / Deny

Actions taken by the firewall to permit or block traffic.



Port

A communication endpoint used by services and applications.



Logging

Records traffic and events for monitoring, troubleshooting and compliance.

SECURITY BENEFITS



Blocks Unauthorized Access

Prevents unauthorized users from accessing your network.



Prevents Attacks

Stops malware, exploits and other cyber threats.



Enforces Policies

Ensures network usage follows security and compliance policies.



Protects Data

Reduces the risk of data breaches and information leaks.



Improves Visibility

Provides insights into traffic, users and applications.



Ensures Continuity

Keeps networks and services secure and running smoothly.

50 NETWORKING DEVICES EXPLAINED

Essential Devices That Power Modern Networks

1 Router  Connects different networks and routes data packets.	2 Switch  Connects devices in a LAN and forwards data.	3 Hub  Basic multipoint repeater; sends data to all ports.	4 Modem  Modulates/demodulates signals for ISP connection.	5 Wireless Access Point  Provides Wi-Fi access and connects to wired network.	6 Firewall  Monitors traffic and protects network from threats.	7 Network Bridge  Connects two LANs and filters traffic.	8 Repeater  Regenerates signals to extend network range.	9 Network Gateway  Connects different networks using different protocols.	10 Load Balancer  Distributes network traffic across multiple servers.
11 Proxy Server  Acts as an intermediary for client requests.	12 DHCP Server  Automatically assigns IP addresses to devices.	13 DNS Server  Translates domain names into IP addresses.	14 File Server  Stores and shares files across the network.	15 Print Server  Manages network printers and print queues.	16 Mail Server  Handles email sending, receiving and storage.	17 Web Server  Hosts websites and web applications.	18 Application Server  Hosts applications and business logic.	19 Database Server  Stores and manages network databases.	20 VPN Server  Provides secure remote access over the internet.
21 NAS (Network Storage)  Centralized storage accessible over the network.	22 SAN (Storage Area Network)  High-speed network for storage devices.	23 Network Interface Card (NIC)  Connects a device to the network.	24 Media Converter  Converts media types (e.g., fiber to copper).	25 SFP Module  Pluggable module for fiber or copper connections.	26 PoE Injector  Adds Power over Ethernet to network cables.	27 PoE Switch  Switch with built-in PoE power for devices.	28 Network Tap  Monitors network traffic without interruption.	29 IDS (Intrusion Detection System)  Monitors network for malicious activity.	30 IPS (Intrusion Prevention System)  Prevents and blocks malicious traffic.
31 UTM (Unified Threat Management)  Combines firewall, antivirus, IPS and more.	32 SD-WAN Device  Optimizes WAN performance using software-defined tech.	33 WAN Optimizer  Improves application performance over WAN.	34 Ethernet Extender  Extends Ethernet beyond 100 meters.	35 Wi-Fi Controller  Manages and controls multiple access points.	36 Captive Portal  Controls and authenticates user network access.	37 RADIUS Server  Centralized authentication, authorization and accounting.	38 TACACS+ Server  Provides AAA for device access control.	39 NTP Server  Synchronizes time across network devices.	40 SNMP Manager  Monitors and manages network devices.
41 Network Analyzer  Captures and analyzes network traffic.	42 Packet Broker  Optimizes traffic flow to monitoring tools.	43 DDoS Protection Device  Detects and mitigates DDoS attacks.	44 WAN Router  Routes traffic across wide area networks.	45 Core Switch  High-speed backbone traffic switching.	46 Edge Switch  Connects end devices at the network edge.	47 KVM over IP  Access and control computers over IP network.	48 IoT Gateway  Connects IoT devices to cloud or networks.	49 Bluetooth Gateway  Bridges Bluetooth devices to IP networks.	50 Zigbee Gateway  Connects Zigbee devices to IP networks.



Build Faster Networks
Use the right devices for efficiency and speed.



Improve Security
Protect your network from threats and attacks.



Ensure Reliability
Keep your network stable and always available.



Scale with Confidence
Choose devices that grow with your business.

“The right networking devices build the foundation for a strong, secure and scalable network.”



NETWORKING
WITH ISRAR

TELNET VS SSH

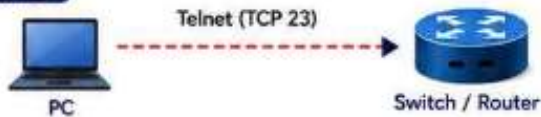
REMOTE ACCESS PROTOCOL COMPARISON



TELNET

What is Telnet?	A protocol used for remote device management.
Port Number	TCP 23
Security	✗ No Encryption ✗ Plain Text Communication
Authentication	✗ Username & Password sent in clear text
Usage	• Legacy Networks • Lab Environments • Testing Purposes

EXAMPLE



RISK

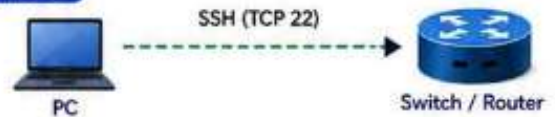
Passwords can be captured by attackers.



SSH

What is SSH?	Secure Shell protocol for encrypted remote management.
Port Number	TCP 22
Security	✓ Encrypted Communication ✓ Secure Authentication
Authentication	✓ Credentials Protected ✓ Public/Private Key Support
Usage	• Enterprise Networks • Data Centers • Production Environments

EXAMPLE



BENEFIT

Protects login credentials and management traffic.

PACKET COMPARISON

TELNET

```
>_ Username: admin
    Password: cisco123
```

✗ Visible in Plain Text

VS

SSH

```
>_ Encrypted Data
    X7#K2@M9!P...
```

✓ Encrypted & Secure

CISCO CONFIGURATION (TELNET)

```
>_ line vty 0 15
    password cisco123
    login
    transport input telnet
```

CISCO CONFIGURATION (SSH)

```
>_ hostname R1
    ip domain-name lab.local
    crypto key generate rsa
    username admin secret Cisco123
    line vty 0 15
    login local
    transport input ssh
```

NETWORK TOPOLOGY



Administrator PC

SSH / Telnet
Remote Access



Cisco Router



Switch



Server

Network



NETWORKING WITH ISRAR
LEARN | PRACTICE | SUCCEED



BUILD SECURE NETWORKS.
PROTECT WHAT MATTERS.

FOLLOW FOR MORE
NETWORKING CONTENT!





BGP STATES

Understanding the Lifecycle of a BGP Peering Session

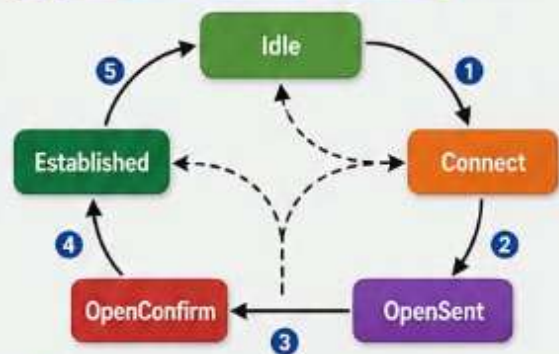


BGP (Border Gateway Protocol) is the routing protocol of the Internet. BGP states represent the different stages of a BGP peer session from start to finish.

1 BGP STATES OVERVIEW

STATE	DESCRIPTION	PURPOSE	TRANSITION
Idle	No TCP connection between peers.	Initial state before starting connection.	Idle → Connect
Connect	TCP connection is being attempted.	Establish TCP connectivity.	Connect → OpenSent
OpenSent	TCP established. BGP OPEN message sent.	Exchange BGP OPEN messages.	OpenSent → OpenConfirm
OpenConfirm	OPEN message received. Waiting for confirmation.	Verify parameters and capabilities.	OpenConfirm → Established (or Idle)
Established	BGP session established. Routes are exchanged.	Exchange routing information.	Established → (Any other state on failure)

2 BGP STATE TRANSITION DIAGRAM



- 1 Start TCP connection.
- 2 TCP connection established.
- 3 BGP OPEN messages exchanged.
- 4 BGP parameters validated.
- 5 BGP session established and routes exchanged.



Any error or failure in the process can cause the session to return to Idle.

3 BGP STATE FLOW (Happy Path)



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4 DESCRIPTION OF EACH STATE

- Idle**
 - No connection exists.
 - Router is waiting or the session is administratively down.
- Connect**
 - Actively trying to establish a TCP connection to the peer.
 - Retries until TCP connection is successful or fails.
- OpenSent**
 - TCP connection is up.
 - BGP OPEN message is sent to the peer.
- OpenConfirm**
 - BGP OPEN message received from peer.
 - Waiting for KEEPALIVE message.
- Established**
 - KEEPALIVE message received.
 - Routes are exchanged and the session is up.

5 COMMON EVENTS & STATE CHANGES

EVENT / CONDITION	RESULTING STATE
Start BGP session	Idle → Connect
TCP connection successful	Connect → OpenSent
BGP OPEN message sent	OpenSent → OpenConfirm
BGP OPEN message received and valid	OpenConfirm → Established
KEEPALIVE message received	Session remains in Established
Error in OPEN message / Capability mismatch	Returns to Idle
TCP connection lost	Returns to Idle

6 KEY POINTS

- ✓ BGP uses TCP port 179.
- ✓ A BGP session must go through all states in order (except on failure).
- ✓ The Established state is the only state where routes are exchanged.
- ✓ Keepalives keep the session alive.
- ✓ Any issue in the process resets the session to Idle.



7 TROUBLESHOOTING TIPS

- ✓ Check TCP connectivity (telnet <peer-ip> 179)
- ✓ Use: show ip bgp summary
- ✓ Review logs for OPEN message errors
- ✓ Ensure matching AS numbers and capabilities
- ✓ Check ACLs, firewalls, and route reachability



BGP is the backbone of the Internet. Understanding its states helps in quick issue identification and resolution.



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EIGRP vs OSPF COMPARISON LAB

COMPARE AND ANALYZE EIGRP AND OSPF IN A SAME NETWORK



OBJECTIVE

Compare EIGRP and OSPF in terms of routing table, convergence time, bandwidth usage and administrative distance.

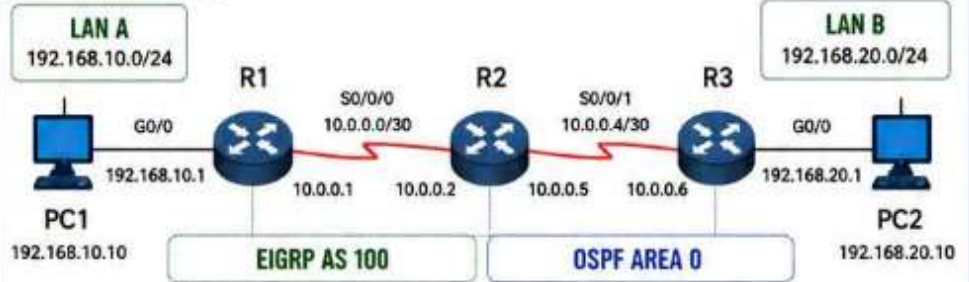


TASKS

1. Configure OSPF in Area 0.
2. Configure EIGRP AS 100.
3. Advertise all networks.
4. Compare routing tables.
5. Test convergence.
6. Compare administrative distance.
7. Analyze bandwidth usage.



TOPOLOGY



IP ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Description
R1	G0/0	192.168.10.1	255.255.255.0	LAN A Gateway
R1	S0/0/0	10.0.0.1	255.255.255.252	To R2 (EIGRP)
R2	S0/0/0	10.0.0.2	255.255.255.252	To R1 (EIGRP)
R2	S0/0/1	10.0.0.5	255.255.255.252	To R3 (OSPF)
R3	S0/0/1	10.0.0.6	255.255.255.252	To R2 (OSPF)
R3	G0/0	192.168.20.1	255.255.255.0	LAN B Gateway
PC1	NIC	192.168.10.10	255.255.255.0	GW: 192.168.10.1
PC2	NIC	192.168.20.10	255.255.255.0	GW: 192.168.20.1



EIGRP CONFIGURATION (R1 & R2)

```
1 enable
2 conf t
3 router eigrp 100
4 no auto-summary
5 network 192.168.10.0 0.0.0.255
6 network 10.0.0.0 0.0.0.3
7 end
```



OSPF CONFIGURATION (R2 & R3)

```
1 enable
2 conf t
3 router ospf 1
4 network 10.0.0.4 0.0.0.3 area 0
5 network 192.168.20.0 0.0.0.255 area 0
6 end
```



VERIFICATION COMMANDS

- show ip route
- show ip protocols
- show ip eigrp neighbors
- show ip ospf neighbor
- show ip cef
- ping 192.168.20.10 (from PC1)
- ping 192.168.10.10 (from PC2)



These commands help you compare routing, convergence and protocol behavior.

EIGRP vs OSPF COMPARISON

Feature	EIGRP	OSPF
Type	Advanced Distance Vector	Link State
Update	Partial (Triggered)	Full (LSA based)
Metric	Bandwidth, Delay, Reliability, Load, MTU	Cost (Bandwidth)
Convergence	Very Fast	Fast
Administrative Distance	90 (Internal)	110 (Internal)
Classless Routing	Yes	Yes
VLSM / CIDR	Yes	Yes
Neighbor Discovery	Hello Packets	Hello Packets
Resource Usage	Lower	Higher (LSA flooding)
Best Use	Smaller to Medium Networks	Large & Scalable Networks

HOW TO TEST CONVERGENCE

1. Start a continuous ping from PC1 to PC2.
2. Shutdown the link between R2 and R3.
3. Note the time when ping starts failing.
4. Bring the link up again.
5. Note the time when ping starts succeeding.



Compare the recovery time (EIGRP vs OSPF).



CHALLENGE QUESTION



If the bandwidth of the link between R1 and R2 is reduced to 64 kbps, which protocol (EIGRP or OSPF) will adapt faster to the change? Why?

How will the routing table entries be different on R2 for both protocols?



Comment your answer below!



QUICK TEST



What is the default AD of EIGRP?



What type of routing is OSPF?



Which protocol uses LSA?



Which protocol has faster convergence?



Which command shows routing protocols?



SUBNETTING vs SUPERNETTING

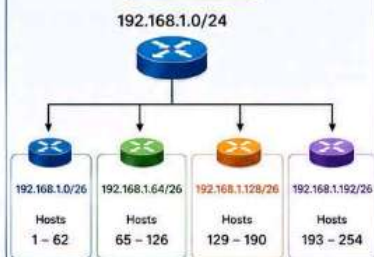
Understanding the Opposite Concepts



SUBNETTING

Subnetting is the process of dividing one large network into multiple smaller networks.

HOW IT WORKS



EXAMPLE

Original Network:
192.168.1.0/24

↓

Divided into 4 Subnets:
/26 (255.255.255.192)

- ✓ More Subnets
- ✓ Smaller Network Size
- ✓ Used Inside Networks

SUBNETTING SUMMARY

Purpose	Divide one network into many
Prefix	Becomes larger (e.g. /24 → /26)
Result	More subnets, smaller size
Usage	Internal network segmentation
Benefit	Better security, efficient IP usage

OPPOSITES



EASY RULE TO REMEMBER

SUBNETTING = ONE NETWORK INTO MANY

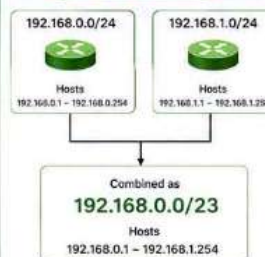
SUPERNETTING = MANY NETWORKS INTO ONE



SUPERNETTING

Supernetting is the process of combining multiple smaller, continuous networks into one larger network.

HOW IT WORKS



EXAMPLE

Networks:

- 192.168.0.0/24
- 192.168.1.0/24

↓

Combined as:
192.168.0.0/23
(255.255.254.0)

- ✓ Fewer Routes
- ✓ Larger Network Size
- ✓ Used Between Routers / ISPs

SUPERNETTING SUMMARY

Purpose	Combine many networks into one
Prefix	Becomes smaller (e.g. /24 + /24 → /23)
Result	Fewer routes, larger size
Usage	Route aggregation between routers / ISPs
Benefit	Reduces routing table size, improves efficiency

BENEFITS

- Better Security**
Isolate departments and reduce threats.
- Efficient IP Usage**
Use IP addresses in a smart way.
- Smaller Routing Table**
Supernetting reduces routing table size.
- Improved Performance**
Less traffic, faster network.
- Scalable Network**
Easier to manage and expand.



SCAN ME

KEY PROTOCOLS OF NETWORK

Protocol		Full Form	Function (Simplified)
HTTP		HyperText Transfer Protocol	Used to load web pages from servers.
HTTPS		HTTP Secure	Encrypts data for safe browsing and shopping online.
FTP		File Transfer Protocol	Used to upload/download files over the internet.
TCP		Transmission Control Protocol	Ensures reliable data delivery between devices.
UDP		User Datagram Protocol	Fast but less reliable than TCP (e.g., video streaming).
IP		Internet Protocol	Helps devices locate and talk to each other online.
SMTP		Simple Mail Transfer Protocol	Sends emails from your computer to the mail server.
POP3		Post Office Protocol v3	Downloads emails from server and deletes them there.
IMAP		Internet Message Access Protocol	Reads emails directly on the server (syncs across devices).
DNS		Domain Name System	Converts website names (like google.com) to IP addresses.
DHCP		Dynamic Host Configuration Protocol	Automatically gives IP addresses to devices.
SSH		Secure Shell	Secure way to access remote computers or servers.
TELNET		Terminal Network	Remote login like SSH but not secure.
SNMP		Simple Network Management Protocol	Used to monitor and manage network devices.





NETWORKING
WITH ISRAR

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SWITCH CONFIGURATION COMMANDS



Essential Cisco Switch Commands for CCNA



SECURE
YOUR NETWORK



OPTIMIZE
PERFORMANCE



ORGANIZE
WITH VLANs



CONTROL
ACCESS



VERIFY
EVERY STEP

BASIC SWITCH SETUP

```
Switch> enable ! Enter privileged mode
Switch# configure terminal ! Global configuration mode
Switch(config)# hostname SW1 ! Set hostname
SW1(config)# enable secret Cisco123 ! Secure privileged mode
SW1(config)# service password-encryption ! Encrypt passwords
SW1(config)# banner motd #
Unauthorized Access Prohibited # ! Set MOTD banner
SW1(config)# no ip domain-lookup ! Disable DNS lookup
SW1(config)# interface vlan 1 ! Management interface
SW1(config-if)# ip address 192.168.1.2 255.255.255.0
SW1(config-if)# no shutdown
SW1(config-if)# exit
SW1(config)# ip default-gateway 192.168.1.1 ! Set default gateway

SW1(config)# line console 0 ! Console access
SW1(config-line)# password Cisco123
SW1(config-line)# login
SW1(config-line)# logging synchronous
SW1(config-line)# exec-timeout 10 0
SW1(config-line)# exit

SW1(config)# line vty 0 15 ! Telnet/SSH access
SW1(config-line)# password Cisco123
SW1(config-line)# login
SW1(config-line)# transport input telnet
SW1(config-line)# exit

SW1(config)# username admin privilege 15 Cisco123 ! Local user

SW1(config)# ip domain-name network.local ! Set domain name

SW1(config)# crypto key generate rsa
2048 ! Generate RSA keys

SW1(config)# ip ssh version 2 ! Enable SSH

SW1# copy running-config startup-config ! Save configuration
```

VLAN CONFIGURATION

```
SW1(config)# vlan 10
SW1(config-vlan)# name SALES
SW1(config-vlan)# exit

SW1(config)# vlan 20
SW1(config-vlan)# name HR
SW1(config-vlan)# exit

SW1(config)# vlan 30
SW1(config-vlan)# name IT
SW1(config-vlan)# exit

SW1(config)# show vlan brief ! Verify VLANs
```

ACCESS PORT CONFIGURATION

```
SW1(config)# interface FastEthernet0/1
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 10
SW1(config-if)# description PC - Sales
SW1(config-if)# spanning-tree portfast
SW1(config-if)# no shutdown
SW1(config-if)# exit

SW1(config)# interface range fa0/2 - 5
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 20
SW1(config-if-range)# exit
```

TRUNK PORT CONFIGURATION

```
SW1(config)# interface GigabitEthernet0/1
SW1(config-if)# switchport mode trunk
SW1(config-if)# switchport trunk allowed vlan
10,20,30

SW1(config-if)# description Uplink-to-Router
SW1(config-if)# no shutdown
SW1(config-if)# exit

SW1(config)# interface range gi0/2 - 3
SW1(config-if-range)# switchport mode trunk
SW1(config-if-range)# switchport trunk allowed
vlan 10,20,30
SW1(config-if-range)# exit
```

VERIFICATION COMMANDS

- show running-config
- show startup-config
- show ip interface brief
- show vlan brief
- show interfaces status
- show interfaces switchport
- show interfaces trunk
- show mac address-table
- show mac address-table dynamic
- show spanning-tree
- show cdp neighbors
- show cdp neighbors detail
- show version
- show flash
- show inventory
- show environment
- show interfaces description
- show interface vlan 1
- show interfaces fa0/1
- show port-security
- show port-security interface fa0/1
- show ip ssh
- show ssh
- show users
- show logging
- show arp
- show clock
- show processes cpu
- show memory statistics
- show history

VLAN QUICK REFERENCE

VLAN	NAME	PURPOSE
10	SALES	Sales Department
20	HR	Human Resources
30	IT	IT Department

SECURITY BEST PRACTICES



Use enable secret
instead of enable
password



Use strong
passwords and
local users



Enable SSHv2
and disable
Telnet



Encrypt all
passwords



Use banners to warn
unauthorized
users



Save
configuration

COMMAND MODE SHORTCUT

```
Switch> User EXEC Mode
Switch# Privileged EXEC Mode
Switch(config)# Global Configuration Mode
Switch(config-if)# Interface Configuration Mode
Switch(config-line)# Line Configuration Mode
```



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NETWORKING CONTENT!



SHOW IP ROUTE

CISCO ROUTER COMMAND

The **show ip route** command displays the IP routing table of the router. It shows all known routes and how they were learned.



MAIN FUNCTIONS

1

View Routing Table

Displays the IP routing table stored in the router.

Command: **show ip route**

2

View Connected Routes

Shows all directly connected networks.

Command: **show ip route**
(Look for codes 'C' and 'L')

3

View Static Routes

Displays manually configured static routes.

Command: **show ip route**
(Look for codes 'S')

4

View Dynamic Routes

Shows routes learned from routing protocols (RIP, OSPF, EIGRP, BGP).

Command: **show ip route**
(Look for codes 'D', 'O', 'R', 'B', 'EX')

5

View Default Route

Displays the default gateway route (0.0.0.0/0).

Command: **show ip route**
(Look for code '*' or 'S*')

6

View Route Source

Shows how each route was learned and its administrative distance.

Command: **show ip route**

EXAMPLE OUTPUT

Router# show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP,
M - mobile, B - BGP, D - EIGRP, EX - EIGRP external,
O - OSPF, IA - OSPF inter area
* - candidate default, U - per-user static route,
o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

```
S* 0.0.0.0/0 [1/0] via 192.168.1.1
C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
L 192.168.1.1/32 is directly connected, GigabitEthernet0/0
C 10.1.1.0/24 is directly connected, GigabitEthernet0/1
L 10.1.1.1/32 is directly connected, GigabitEthernet0/1
D 172.16.0.0/16 [90/2172416] via 10.1.1.2, 00:12:34, Gi0/1
O 192.168.2.0/24 [110/2] via 10.1.1.2, 00:05:21, Gi0/1
```

ROUTE CODE LEGEND

Code	Meaning	Description	Admin Distance
C	Connected	Directly connected network	0
L	Local	Local IP address of the router interface	0
S	Static	Manually configured static route	1
S*	Static Default	Default static route (gateway of last resort)	1
D	EIGRP	Route learned via EIGRP	90 (Internal)
EX	EIGRP External	External route learned via EIGRP	170
O	OSPF	Route learned via OSPF	110
IA	OSPF Inter Area	Inter-area route learned via OSPF	110
R	RIP	Route learned via RIP	120
B	BGP	Route learned via BGP	20
*	Candidate Default	Potential default route	-
U	Per-user Static	Per-user static route	-
M	Mobile	Mobile route	-
o	ODR	OSPF On Demand Route	-
P	Periodic Downloaded	Periodic downloaded static route	-

COMMAND SYNTAX

show ip route

Optional Filters:

- **show ip route <network>**
- **show ip route <network> <mask>**
- **show ip route <protocol>**
- **show ip route <prefix-length>**

COMMON USES



Verify Connectivity

Check if a destination network is reachable and via which path.



Troubleshoot Routing Issues

Identify missing routes, wrong next hops, or routing loops.



Verify Default Gateway

Ensure a valid default route exists for internet access.



Monitor Dynamic Routing

Confirm routes learned from routing protocols.



Analyze Route Preference

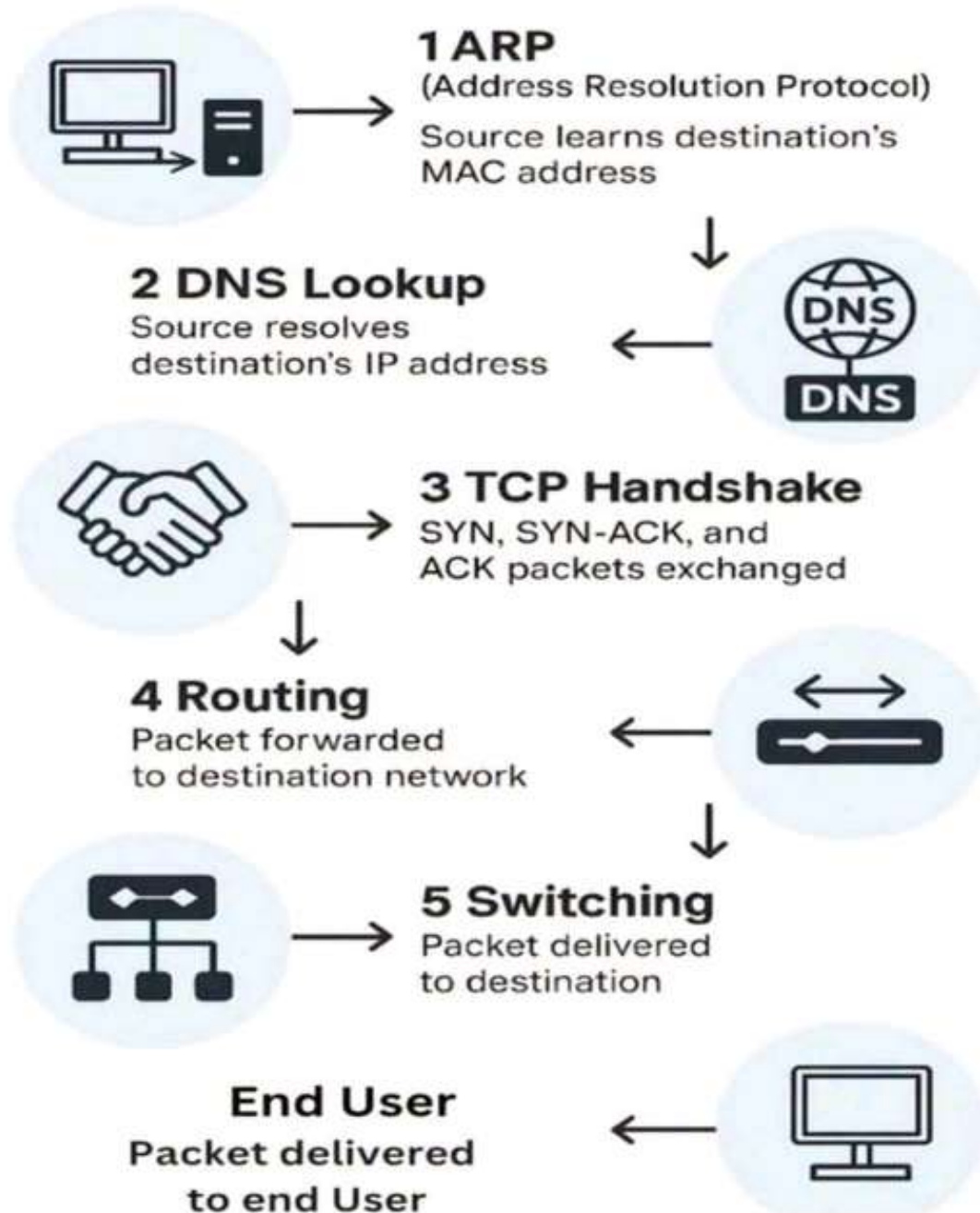
Compare administrative distance and route selection.



TIPS

- Use **show ip route | include <network>** to filter specific routes.
- Use **show ip route summary** to get a quick summary of all routes.

Packet Flow in Network



2-BIT SUBNETTING

STEP BY STEP EXAMPLE



WHAT IS 2-BIT SUBNETTING?

2-bit subnetting means we borrow 2 bits from the host portion of the network address to create smaller subnets.

EXAMPLE NETWORK

Network Address : 192.168.1.0/24

Default Subnet Mask : 255.255.255.0

We borrow 2 bits from the host portion.



STEP 1: NEW SUBNET MASK

$$/24 + 2 \text{ (borrowed bits)} = /26$$

New Subnet Mask : 255.255.255.192

Binary :

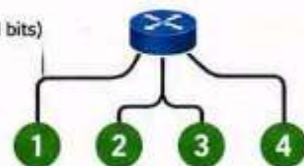
255.255.255.0 : 11111111.11111111.11111111.00000000

255.255.255.192 : 11111111.11111111.11111111.11000011

STEP 2: NUMBER OF SUBNETS

Formula : 2^n (where n = borrowed bits)

$$2^2 = 4 \text{ Subnets}$$



STEP 3: HOSTS PER SUBNET

Remaining Host Bits = 8 - 2 = 6

Formula : $2^{\text{Host Bits}} - 2$

$$2^6 - 2 = 64 - 2 = 62 \text{ Hosts}$$



STEP 4: BLOCK SIZE (INCREMENT)

Block Size = 256 - New Subnet Mask (Last Octet)

$$256 - 192 = 64$$

Subnet Increment = 64



SUBNET TABLE (2-BIT SUBNETTING)

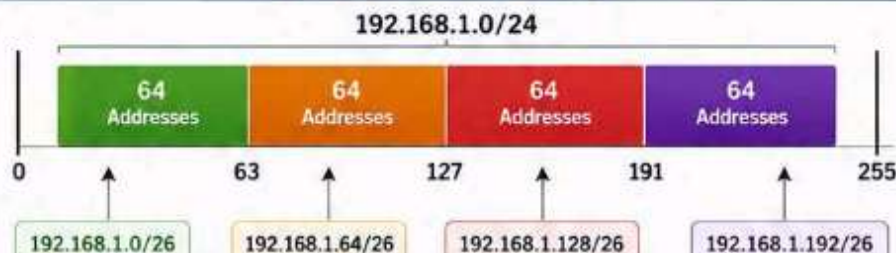
Subnet No.	Network ID (/26)	First Host	Last Host	Broadcast Address	Total Hosts
1	192.168.1.0/26	192.168.1.1	192.168.1.62	192.168.1.63	62
2	192.168.1.64/26	192.168.1.65	192.168.1.126	192.168.1.127	62
3	192.168.1.128/26	192.168.1.129	192.168.1.190	192.168.1.191	62
4	192.168.1.192/26	192.168.1.193	192.168.1.254	192.168.1.255	62

QUICK SUMMARY

- ✓ Borrowed Bits : 2
- ✓ New Prefix : /26
- ✓ New Subnet Mask : 255.255.255.192
- ✓ Number of Subnets : 4
- ✓ Hosts per Subnet : 62
- ✓ Subnet Increment : 64



VISUAL REPRESENTATION



Remember

More borrowed bits = More subnets
Less host bits = Fewer hosts per subnet

Follow MR IT & Networking
for more CCNA tips and tricks!



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& NETWORKING

LEARN TODAY | LEAD TOMORROW

BGP

BORDER GATEWAY PROTOCOL



THE PROTOCOL THAT POWERS THE INTERNET

1 BGP NEIGHBOR ESTABLISHMENT



```
R1# show ip bgp summary
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	State/PfxRcd
203.0.113.2	4	65002	52	47	12	Established 25



PURPOSE

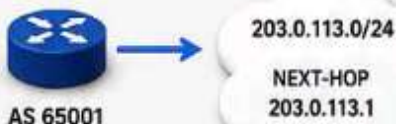
Establish a TCP session (port 179) and exchange BGP messages.



VERIFY

show ip bgp summary

2 BGP ROUTE ADVERTISEMENT



```
R1# show ip bgp
```

Network	Next Hop	Path
*> 203.0.113.0/24	203.0.113.1	65001



PURPOSE

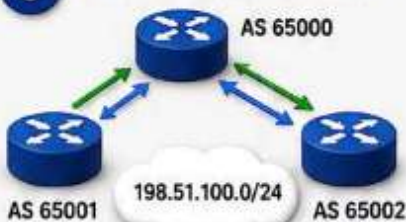
Advertise network prefixes to BGP neighbors.



VERIFY

show ip bgp

3 BGP PATH SELECTION



```
R1# show ip bgp 198.51.100.0
```

BGP routing table entry for 198.51.100.0/24

Paths: (2 available, best #1)

65001 65000

65002 65000

Best path: 65001 65000



PURPOSE

Select the best path using BGP best path algorithm.



VERIFY

show ip bgp <prefix>

4 BGP POLICY CONTROL



```
R1(config)# ip prefix-list ALLOW permit 203.0.113.0/24
```

```
R1(config)# route-map OUT deny 10
```

```
R1(config-route-map)# match ip address prefix-list ALLOW
```

```
R1(config-route-map)# set community no-export
```

```
R1(config)# router bgp 65001
```

```
R1(config-router)# neighbor 203.0.113.2 route-map OUT out
```



PURPOSE

Control what routes to advertise or accept using policies.



VERIFY

show ip bgp | include <prefix>

5 BGP MULTI-HOMING



```
R1# show ip bgp
```

Network	Next Hop	Path
*> 8.8.8.0/24	192.0.2.1	65010
* 8.8.8.0/24	198.51.100.1	65020

Best path: 192.0.2.1 via 65010



PURPOSE

Maintain multiple connections to different AS for redundancy.



VERIFY

show ip bgp

KEY TAKEAWAYS



BGP is the backbone of the Internet.



It exchanges routes between ASes.



Best path selection ensures efficiency.



Policies control routing decisions.



Scalability & reliability come from good design.



DESIGN SMART. ADVERTISE WISELY. KEEP THE INTERNET RUNNING.

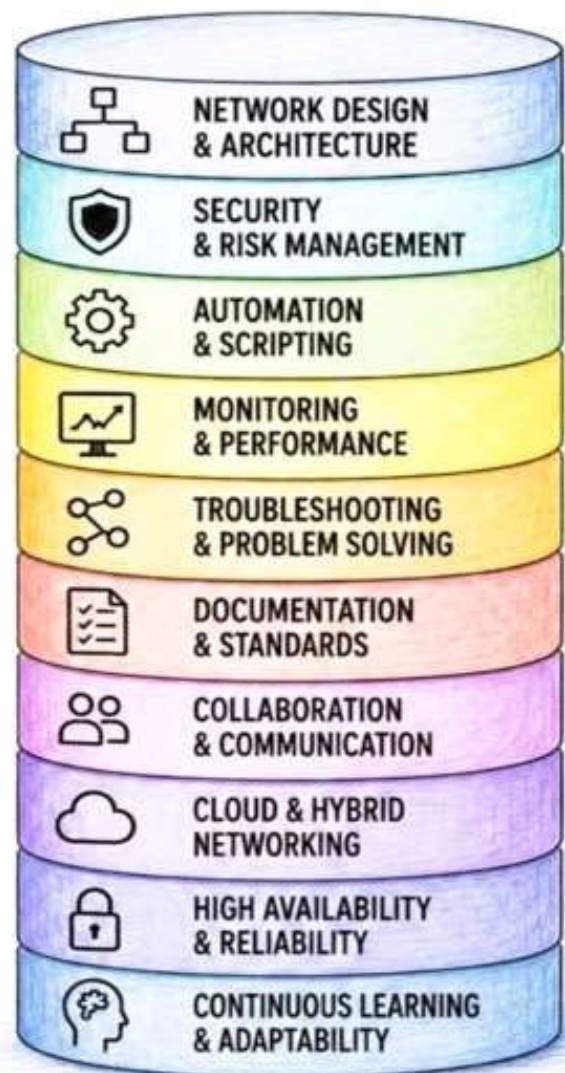
www.thenetworkdna.com

NETWORK ENGINEER

What people
think a network engineer
does



What a network engineer
really does



STP CONFIGURATION LAB

CONFIGURE AND VERIFY SPANNING TREE PROTOCOL

OBJECTIVE

Configure and verify Spanning Tree Protocol (STP) to prevent Layer 2 loops and ensure network stability.

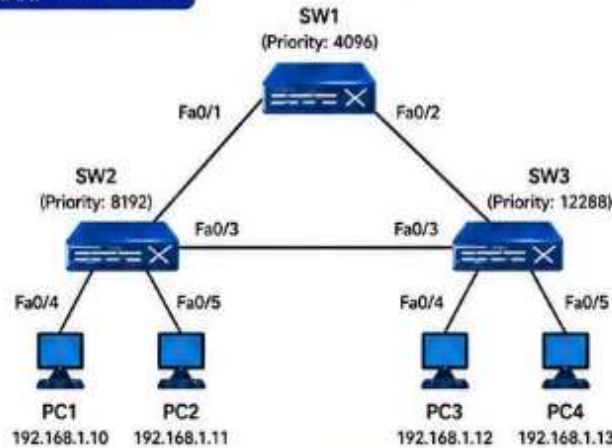


TASKS

1. Configure basic STP on all switches.
2. Identify the root bridge.
3. Verify port roles and states.
4. Block redundant links to prevent loops.
5. Test connectivity between PCs.
6. Modify STP priority and observe changes.



TOPOLOGY



STP SUMMARY

- Prevents Layer 2 loops
- Blocks redundant paths
- Ensures one active path between any two devices
- Provides backup links in case of failure

VERIFICATION CHECKS

- Root bridge is SW1.
- SW2 Fa0/1 and SW3 Fa0/2 are Root Ports.
- SW2 Fa0/3 is Designated Port (Forwarding).
- SW3 Fa0/3 is Alternate Port (Blocking).
- All access ports are Forwarding.



EXPECTED STP ROLES AND STATES

Switch	Port	Role	State	Description
SW1	All	Root	Forwarding	SW1 is the Root Bridge (lowest priority)
SW2	Fa0/1	Root Port	Forwarding	Best path to root via SW1
SW2	Fa0/3	Designated Port	Forwarding	Designated on the segment
SW3	Fa0/2	Root Port	Forwarding	Best path to root via SW1
SW3	Fa0/3	Alternate Port	Blocking	Redundant link (blocked by STP)
All	Fa0/4, Fa0/5	Designated Port	Forwarding	Access ports to PCs

SW1 - CONFIGURATION (ROOT BRIDGE)

```
SW1# configure terminal
SW1(config)# spanning-tree mode pvst
SW1(config)# spanning-tree vlan 1 root primary
SW1(config)# end
SW1# write memory
```

SW2 - CONFIGURATION

```
SW2# configure terminal
SW2(config)# spanning-tree mode pvst
SW2(config)# end
SW2# write memory
```

SW3 - CONFIGURATION

```
SW3# configure terminal
SW3(config)# spanning-tree mode pvst
SW3(config)# end
SW3# write memory
```

IMPORTANT SHOW COMMANDS

- show spanning-tree
- show spanning-tree vlan 1
- show spanning-tree summary
- show spanning-tree interface fa0/1



These commands help you verify root bridge, port roles and port states.

STP ROOT BRIDGE ELECTION

1. The switch with the lowest Bridge ID (Priority + MAC Address) becomes the Root Bridge.
2. Default priority is 32768.
3. Lower priority value = higher preference.

Bridge ID Comparison

Switch	Priority	MAC Address (Example)	Bridge ID
SW1	4096	0001.96AA.AA01	4096.0001.96AA.AA01
SW2	8192	0001.96AA.AA02	8192.0001.96AA.AA02
SW3	12288	0001.96AA.AA03	12288.0001.96AA.AA03

★ SW1 has the lowest Bridge ID → SW1 is the Root Bridge.

STP PORT STATES

STP places each port in one of the following states:

- Blocking**: Does not forward frames, only listens for BPDUs.
- Listening**: Listens for BPDUs, not learning MAC addresses.
- Learning**: Learns MAC addresses, does not forward frames.
- Forwarding**: Forwards frames and learns MAC addresses.

STP transitions through these states before becoming Forwarding.



CHALLENGE QUESTION

If the priority of SW3 is changed to 2048, what will happen?

- A. SW3 becomes the Root Bridge
- B. SW1 remains the Root Bridge
- C. SW2 becomes the Root Bridge
- D. All switches will become Root Bridges

What will be the new port roles and states?

Comment your answer below!

QUICK TIPS

- Always set a known Root Bridge for predictable behavior.
- Never disable STP in production networks.
- Monitor STP status regularly using show commands.
- Document your changes and configurations.



Types of Network

Date : 17/06/26

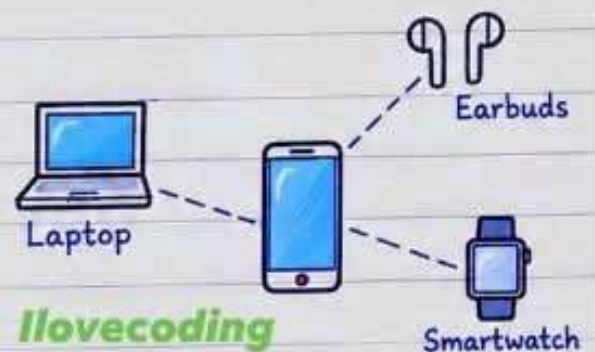
Page No. : 1

A **Computer Network** is a group of connected computers and devices that **share data and resources**.

1. Types (Based on Size)

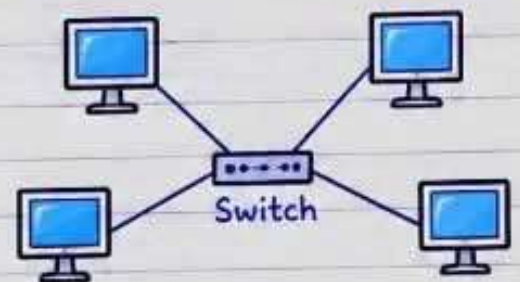
① **PAN** (Personal Area Network)

Very small network around a person (e.g. Bluetooth).



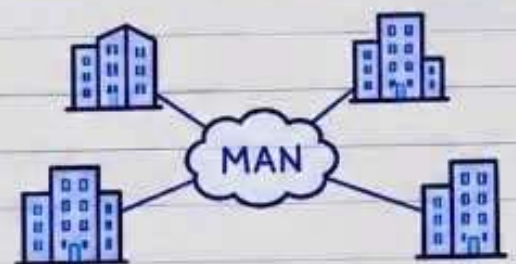
② **LAN** (Local Area Network)

Covers small area like a room, building or office.



③ **MAN** (Metropolitan Area Network)

Covers city or large campus.



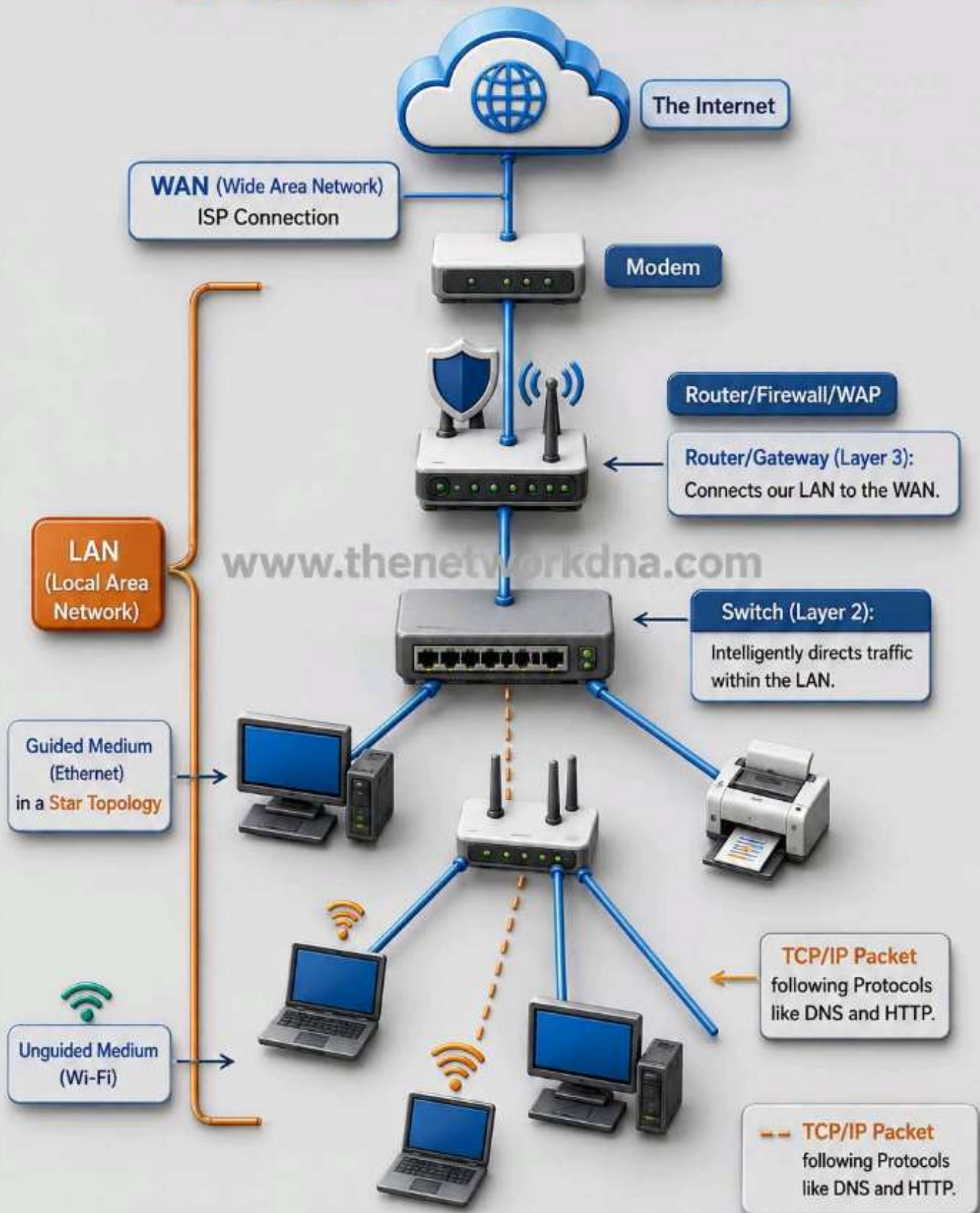
④ **WAN** (Wide Area Network)

Covers large geographical areas (country or worldwide).



From Blueprint to Backbone:

A Small Office Network



Reliable
Connectivity



Secure
Network



Efficient Traffic
Management



Scalable for
Future Growth

IP ADDRESSING & SUBNETTING

THE FOUNDATION OF EFFICIENT NETWORK COMMUNICATION

IP ADDRESSING BASICS

An IP address is a unique identifier assigned to each device on a network so they can communicate.

IPv4 ADDRESS FORMAT

192 . 168 . 1 . 10

1st Octet (0-255) 2nd Octet (0-255) 3rd Octet (0-255) 4th Octet (0-255)

TYPES OF IP ADDRESSES

- **Network Address** – Identifies the network
- **Host Address** – Identifies a device on the network
- **Broadcast Address** – Reaches all devices on the network



IP address = Network ID + Host ID



Every device needs an IP to communicate.



SUBNETTING BASICS

Subnetting divides a large network into smaller, more efficient sub-networks.

EXAMPLE: 192.168.1.0/24

/24 means 24 bits for Network ID and 8 bits for Host ID

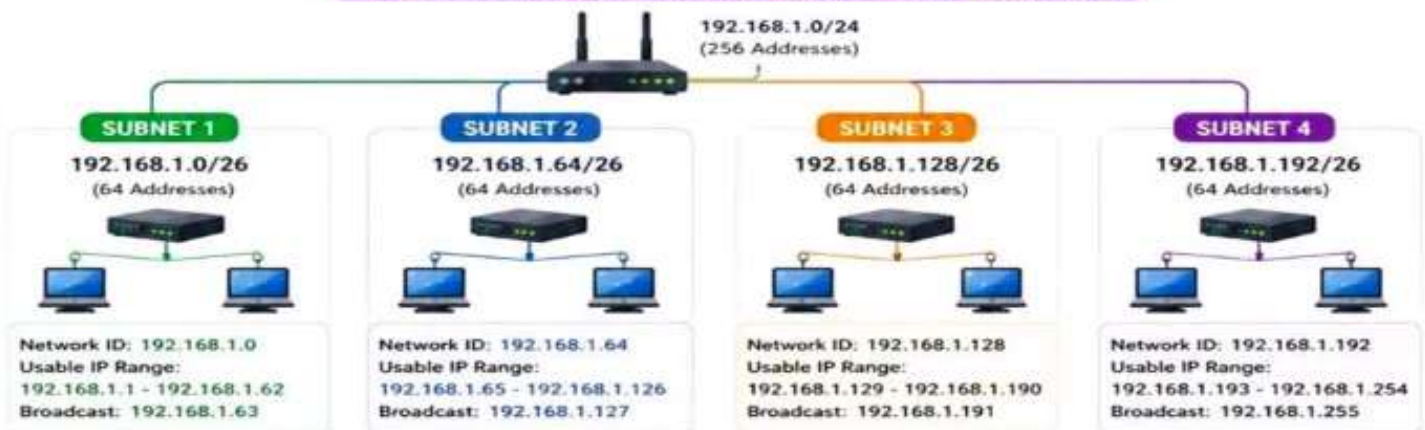
Network ID (24 bits)

Host ID (8 bits)

SUBNETTING BENEFITS

- ✓ Efficient IP address utilization
- ✓ Improved network performance
- ✓ Enhanced security
- ✓ Easier network management

EXAMPLE: SUBNETTING 192.168.1.0/24 INTO 4 SUBNETS



KEY TAKEAWAY



IP addressing identifies devices. Subnetting organizes networks. Together, they build the backbone of every well-designed network.

KEY TERMS

- **IP Address** – Unique address of a device
- **Subnet Mask** – Determines network & host portions
- **Network ID** – The address of the network
- **Host ID** – The address of a device on the network
- **Broadcast** – Sends to all devices in the network

CIDR NOTATION EXPLAINED

192.168.1. 0/24

Network Address

Prefix Length (24 bits)

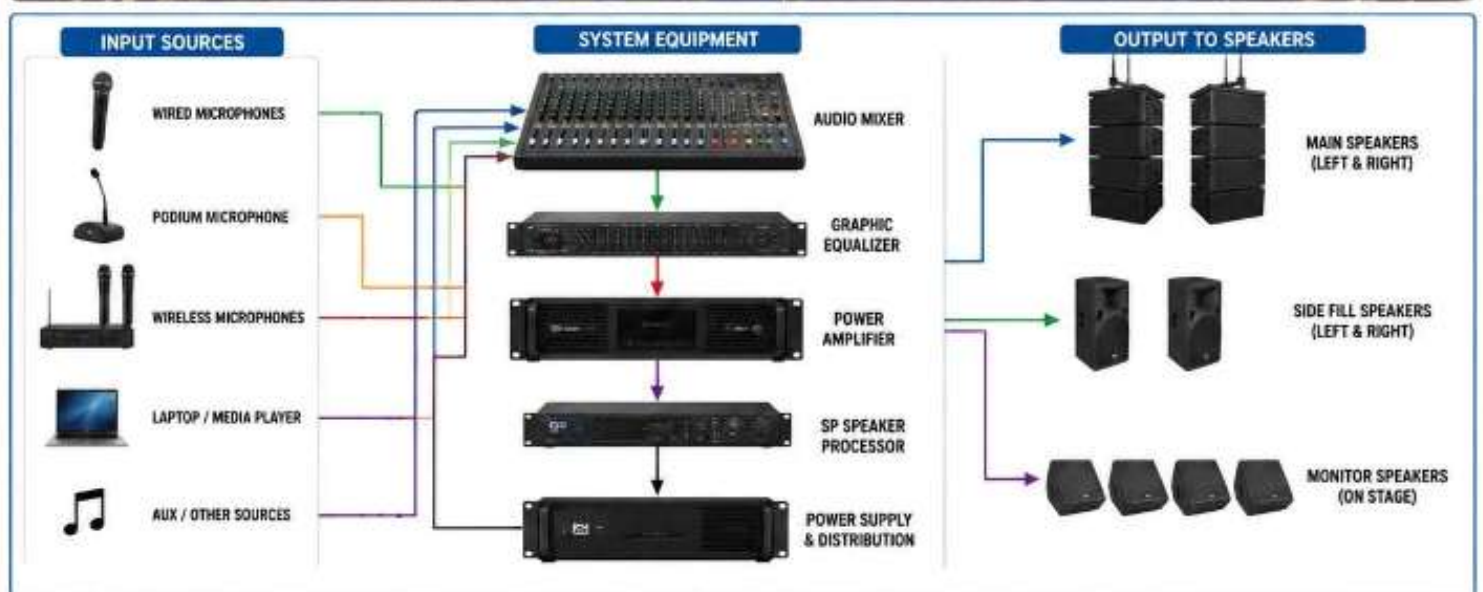


Well-planned IP addressing and subnetting lead to scalable, secure and efficient networks.

PA SYSTEM ARRANGEMENT

IN THE AUDITORIUM

CLEAR SOUND • WIDER COVERAGE • BETTER EXPERIENCE



CLEAR VOICE
INTELLIGIBILITY



EVEN SOUND
COVERAGE



RELIABLE &
SAFE SYSTEM



PROFESSIONAL
INSTALLATION



BETTER EXPERIENCE
FOR EVERYONE

GOOD SYSTEM • GOOD SOUND • SUCCESSFUL EVENT

IPV4 SUBNETTING

FORMULAS



1. NUMBER OF SUBNETS

$$2^n$$

n = number of borrowed bits

Example: If you borrow 3 bits
Subnets = $2^3 = 8$



2. NUMBER OF HOSTS PER SUBNET

$$2^h - 2$$

h = number of host bits remaining

-2 is because one address is the network address and one is the broadcast address.



3. BLOCK SIZE (SUBNET INCREMENT)

256 – Subnet Mask Value

This tells us the increment between subnet network addresses.

Example: Mask 255.255.255.224
Block size = $256 - 224 = 32$

EXAMPLE (CLASS C: 192.168.1.0/27)

	Borrowed bits (n)	= 3
	Number of Subnets	= $2^3 = 8$
	Host bits (h)	= 5
	Hosts per Subnet	= $2^5 - 2 = 30$
	Block Size	= $256 - 224 = 32$
	Subnet Mask	= 255.255.255.224
	Prefix Length	= /27

SUBNETS IN 192.168.1.0/27

Subnet #	Network Address	Usable Host Range	Broadcast Address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95
4	192.168.1.96	192.168.1.97 – 192.168.1.126	192.168.1.127
5	192.168.1.128	192.168.1.129 – 192.168.1.158	192.168.1.159
6	192.168.1.160	192.168.1.161 – 192.168.1.190	192.168.1.191
7	192.168.1.192	192.168.1.193 – 192.168.1.222	192.168.1.223
8	192.168.1.224	192.168.1.225 – 192.168.1.254	192.168.1.255

💡 Block size = 32, so the network addresses increase by 32 each time (0, 32, 64, 96, 128, 160, 192, 224)

QUICK FORMULA SUMMARY



Subnets
= 2^n

n = borrowed bits



Hosts per Subnet
= $2^h - 2$

h = host bits remaining



Block Size
= $256 -$
Mask Value



Prefix Length
= Default Prefix
+ Borrowed Bits

DEFAULT PREFIXES

Class A = /8 (255.0.0.0)
Class B = /16 (255.255.0.0)
Class C = /24 (255.255.255.0)



NETWORKING
WITH ISRAR

LEARN | PRACTICE | SUCCEED

TYPES OF IP ADDRESSES

Different IPs. Different Purpose. **Same Network.**



🔄 Every IP Has a Role in Communication.

1 PUBLIC IP ADDRESS

A public IP address is assigned by an ISP and is used to identify devices on the Internet.

Example:
8.8.8.8

Range:
1.0.0.0 – 223.255.255.255



2 PRIVATE IP ADDRESS

Private IP addresses are used within local networks and are not routable on the Internet.

Examples: 192.168.1.10
10.0.0.5
172.16.0.1

Ranges:
10.0.0.0 – 10.255.255.255
172.16.0.0 – 172.31.255.255
192.168.0.0 – 192.168.255.255



3 LOOPBACK IP ADDRESS

Loopback IP is used for testing and troubleshooting on the local device.

Example:
127.0.0.1

Range:
127.0.0.0 – 127.255.255.255



DEVICE ITSELF

4 LINK-LOCAL IP ADDRESS

Used for communication between devices on the same local network segment when no router is available.

Example (IPv4):
169.254.10.20

Range:
169.254.0.0 – 169.254.255.255



SAME LOCAL LINK

5 MULTICAST IP ADDRESS

Used to send data from one source to multiple destinations.

Example (IPv4):
224.0.0.1

Range:
224.0.0.0 – 239.255.255.255



ONE TO MANY

6 BROADCAST IP ADDRESS

Used to send data to all devices in the same network.

Example (IPv4):
192.168.1.255

Note:
Directed broadcast is rarely used now.



ONE TO ALL

7 ANYCAST IP ADDRESS

The same IP assigned to multiple devices; traffic is routed to the nearest one.

Example:
DNS Root Servers use anycast IPs.



ONE TO NEAREST

8 RESERVED IP ADDRESS

Certain IP addresses are reserved for specific purposes and cannot be used by hosts.

Examples:

- 0.0.0.0 – This Network
- 255.255.255.255 – Limited Broadcast
- 255.255.255.254 – For future use



RESERVED / SPECIAL USE

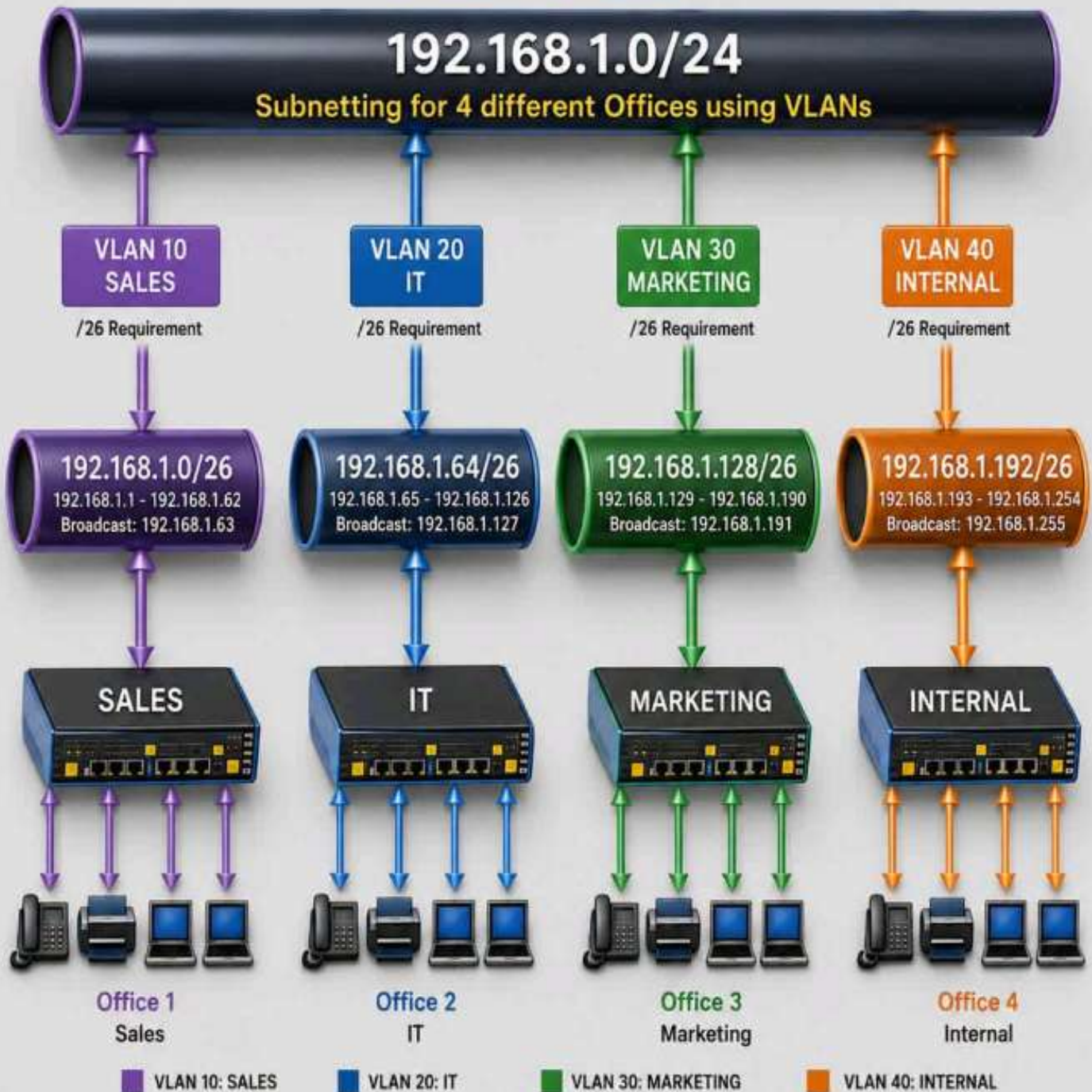


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Network Subnetting Simplify

www.thenetworkdna.com



Each VLAN has its own subnet to reduce broadcast traffic and improve network performance.